

Aspiration-Weighted Influence

Siming Ye*

August 2026

Abstract

Consumers see what others buy and what creators recommend, including things they cannot afford or access. I study a consumer who draws from her baseline choice distribution or from exposure. She repeats the draw until she selects an attainable alternative. Holding the composition of attainable exposure fixed, a larger unattainable share increases the weight on the baseline. Reallocating exposure among unattainable alternatives leaves choice unchanged. Two independence restrictions describe how exposure transfers affect choice probabilities and relative choice. Together with one positive but incomplete response when all exposure is attainable, they characterize the fixed-menu choice rule and uniquely recover its baseline and influence weight. Two informative exposure regimes identify both when the menu, baseline, and influence weight remain fixed. Additional cross-menu restrictions yield a common influence weight and a Luce baseline. A sender application shows how product returns, image value, and the cost of delay determine the mix of attainable and unattainable content.

*Department of Economics, *Georgetown University*. Email: sy677@georgetown.edu. I am deeply grateful to my advisor, Christopher Chambers, for his guidance and support. I thank Yusufcan Masatlioglu, Federico Echenique, John Rust, Garance Genicot, Christopher Turansick, Axel Anderson, Michael Richter, RC Lim, Pietro Ortoleva, Duarte Goncalves, Brice Magdalou, and participants at the 2026 FUR Conference, 2026 BSE Summer Forum Choice and Decision Workshop, and the Behavioral Economics Seminar at Shanghai University of Finance and Economics for helpful comments. Earlier versions of this paper were presented at the 2025 Midwest Economics Association Annual Meeting and a 2025 PhD workshop at Queen Mary University of London. I thank the participants at these events for their valuable feedback. All errors are my own.

1 Introduction

An amateur photographer follows professionals who use lenses far beyond her budget. Her feed also includes lenses she can afford, and the balance between the two kinds of posts can change while her budget stays fixed. Similar differences arise when a home cook watches recipes using ingredients unavailable nearby, or a player sees items locked at her account tier. This paper studies how exposure to attainable and unattainable alternatives affects choice from a fixed menu.

The consumer's baseline choice distribution describes how she chooses from her menu on her own. Exposure records how frequently she sees each alternative, including those outside the menu. On each draw, she uses the baseline with one probability and exposure with the remaining probability. I call the latter probability the influence weight. A draw naming an attainable alternative completes the choice. Following an unattainable draw, she draws again using the same probabilities. The draws are independent, and the menu and baseline remain fixed throughout the process.

The resulting choice distribution is a weighted average of the baseline and the composition of attainable exposure. The weight on exposure rises with its attainable share. For the photographer, a feed containing more unaffordable lenses therefore puts greater weight on her baseline choices, holding the relative frequencies of affordable lenses fixed. Her choice distribution moves toward the baseline whenever the attainable composition differs from it. Replacing posts about one unaffordable lens with posts about another leaves choice unchanged.

I characterize the choice rule through three restrictions on responses to exposure. The first concerns transfers between attainable alternatives. Replacing exposure to one attainable alternative with exposure to another leaves the choice probability of every other attainable alternative unchanged. The second concerns transfers from an unattainable alternative to an attainable one. Such a transfer preserves relative choice between any two other attainable alternatives. Among choices from that pair, each alternative retains its original share whenever the pair has positive choice probability before and after the transfer.

The third restriction requires one positive but incomplete response when all exposure is attainable. For at least one transfer, the recipient's choice probability rises by less than the exposure transferred to it. On a menu with at least three attainable alternatives and at least one alternative outside the menu, the three restrictions characterize the model over all exposure distributions with a positive attainable share. They recover a unique baseline and influence weight that remain fixed as exposure varies. They also imply independence from the allocation of unattainable exposure.

Under the model, two exposure regimes that generate different choice distributions identify the baseline and influence weight when both remain fixed on the same menu. Accounting for each regime’s attainable share and differencing the choice equations eliminates the common baseline. The comparison identifies the influence weight, and either regime then recovers the baseline. A change in the attainable share alone provides this information when the composition of attainable exposure differs from the baseline. Additional regimes impose restrictions using the same recovered parameters. These comparisons require exposure frequencies for attainable alternatives and their share in total exposure.

Across menus, additional conditions compare changes in choice probabilities and relative choice between unexposed alternatives. With positive baseline probabilities and a menu family covering every triple of alternatives, these conditions yield a common influence weight and a Luce baseline. Baseline choice is then proportional to a fixed positive index over alternatives.

An application lets a sender choose exposure for an audience with a common menu. The sender earns product returns from completed choices and image value from displaying unattainable alternatives. When influencing choice raises product returns and unattainable content has positive image value, costless completion leads the optimal attainable share to approach zero as the influence weight approaches one. Expected completion time then diverges. With a positive cost of delay, the optimal attainable share has a strictly positive limit. The cost of completing a choice therefore affects how much attainable content the sender displays.

The paper connects models of stochastic choice and social influence ([Cuhadaroglu, 2017](#); [Chambers et al., 2023](#); [Kashaev et al., 2026](#)) with theories in which alternatives outside the menu affect choice ([Güney and Richter, 2015](#); [Güney et al., 2018](#); [Boissonnet and Ghersengorin, 2026](#)). Section 2 develops these connections. Sections 3 and 4 present the model and its comparative statics. Section 5 gives the characterization, Section 6 studies identification, and Section 7 studies the sender’s content choice.

2 Related Literature

The starting point is the revealed-preference analysis of stochastic choice. A stochastic choice rule records how often each alternative is selected from every feasible set. Luce’s choice axiom and McFadden’s random-utility formulation showed how restrictions on these probabilities can reveal an underlying utility structure ([Luce, 1959](#); [McFadden, 1974](#)). Subsequent work has given randomness

several different foundations. It may reflect a distribution over preference orderings, a perturbation of utility, a limited consideration set, or costly information acquisition ([Gul and Pesendorfer, 2006](#); [Fudenberg et al., 2015](#); [Masatlioglu et al., 2012](#); [Manzini and Mariotti, 2014](#); [Matějka and McKay, 2015](#)). A related line keeps the Luce form and adds a single parameter carrying an economic friction; [Chambers et al. \(2025\)](#) attach a salience weight to each alternative and derive the resulting demand system. The dataset itself can also be the object of study: [Azrieli and Rehbeck \(2026\)](#) characterize what marginal distributions of menus and of choices reveal when conditional choice frequencies go unobserved. Across these approaches, observed variation in the decision problem disciplines an object that the analyst cannot see directly.

A second strand treats an observed choice rule as a mixture of component rules and asks what the mixture reveals. [Suleymanov \(2026\)](#) studies an agent whose choices blend a principal's Luce rule with its own and shows when the degree of alignment between the two is identified from the agent's choices alone. [Chambers et al. \(2024\)](#) take up the companion question across decision problems, characterizing when a joint random choice rule decomposes into independent marginals. Both ask what one observed frequency reveals about the components behind it.

Most of this literature takes the menu as the principal source of variation. Social and digital environments add another observable object: the distribution of behavior, demonstrations, or recommendations that reaches the decision maker. Once choice probabilities are indexed by both a menu and an exposure distribution, the analyst can vary what the consumer sees while holding fixed what she can choose. That distinction is available whenever the same menu is observed under different feeds, peer groups, recommendation policies, or sender strategies.

Choice-theoretic models of social influence study how another person's action, a social norm, or a distribution of peer behavior affects individual choice ([Cuhadaroglu, 2017](#); [Chambers et al., 2023](#); [Borah and Kops, 2019](#); [Volpe, 2025](#)). Social observations may alter preferences, consideration, or both ([Kashaev and Lazzati, 2021](#); [Kashaev et al., 2026](#)). These models commonly give the decision maker and the social source a shared opportunity set, an assumption that fits settings in which peers choose from the same alternatives. A source that broadcasts to an audience with heterogeneous budgets, locations, and access tiers sits outside that arrangement.

The empirical literature on social interactions explains why variation in exposure matters. Peer behavior is often correlated with common shocks, sorting, and endogenous group formation. Those concerns motivate network instruments, random assignment, and experimentally induced information ([Manski, 1993](#); [Sacerdote, 2001](#); [Bursztyн et al., 2014](#); [Blume et al., 2011, 2015](#);

[Bramoullé et al., 2020](#); [Boucher et al., 2024](#)). The exposure considered here arrives before the current choice and need not include that choice, so the immediate problem does not contain the simultaneity that generates the reflection problem. A decomposition problem remains in either case. Frequent selection of an alternative can come from a strong underlying taste or from a strong response to seeing it often, and that literature has met the problem with assigned or otherwise stable variation in what people observe.

An alternative may go unchosen for two quite different reasons. Revealed-attention and random-consideration models take up the first: the alternative is feasible, and what fails is its entry into deliberation ([Masatlioglu et al., 2012](#); [Manzini and Mariotti, 2014](#)). Deliberation and availability come apart in ordinary settings. The photographer may watch an entire review of an expensive lens, remember it vividly, and still be unable to buy it.

Phantom alternatives alter evaluation after they disappear from the feasible set ([Farquhar and Pratkanis, 1993](#)). In aspiration-based choice, an unattainable aspiration guides the decision maker toward a nearby feasible alternative, and restriction-sensitive choice lets a forbidden option redirect demand within the available set ([Güney and Richter, 2015](#); [Güney et al., 2018](#); [Boissonnet and Ghersengorin, 2026](#)). In each of these accounts the identity, similarity, or position of the unavailable object does the work.

The term *aspiration* has a broader meaning in development economics and behavioral economics. Aspirations can affect effort, investment, occupational choice, and the future opportunity set ([Appadurai, 2004](#); [Genicot and Ray, 2017](#); [Dalton et al., 2016](#); [Mookherjee et al., 2010](#)). Reference dependence and status concerns give comparison objects a direct role in utility and demand ([Kőszegi and Rabin, 2006](#); [Veblen, 1899](#); [Bagwell and Bernheim, 1996](#); [Hopkins and Kornienko, 2004](#)). Those theories study how social comparison changes what an individual wants, or can eventually attain, once resources and access have had time to adjust.

Creator commerce brings these threads together, since one content stream reaches viewers with heterogeneous constraints. Social platforms combine product discovery, information, advertising, and affiliation, while budgets and access remain viewer-specific ([Aridor et al., 2024](#)). A creator choosing what to display thus faces an audience whose members differ in what they can act upon, and the returns to a post depend on which viewers it reaches.

3 The Model

3.1 Primitives

Let $\mathcal{X}$ be a finite set of alternatives and let $S \subseteq \mathcal{X}$ be the nonempty menu from which the consumer can choose. Alternatives in S are *attainable*; those outside S are *unattainable*. The menu may depend on prices, income, inventory, location, or an account tier. It may include an outside option representing no purchase. These conditions remain fixed during the choice. Write $\Delta(\mathcal{X})$ and $\Delta(S)$ for the probability distributions on $\mathcal{X}$ and S , respectively.

The consumer also sees behavior or recommendations concerning alternatives in $\mathcal{X}$. Her *exposure distribution* is $q \in \Delta(\mathcal{X})$. For each alternative $x \in \mathcal{X}$, the probability $q(x)$ records its relative frequency in this exposure. Exposure may come from posts, recommendations, or observed purchases, and it may assign positive probability to unattainable alternatives.

The vector $q|_S = (q(x))_{x \in S}$ records exposure to attainable alternatives. Its total mass, $q_S \equiv \sum_{x \in S} q(x)$, is the *attainable share*. When $q_S > 0$, define the *conditional exposure distribution* by $q(x | S) = q(x)/q_S$ for $x \in S$. This distribution describes the composition of attainable exposure. Two feeds can have the same attainable composition while devoting different shares of their posts to attainable alternatives. The remaining share, $1 - q_S$, concerns unattainable alternatives.

The consumer's *baseline choice distribution*, $p_0(\cdot | S) \in \Delta(S)$, describes how she chooses from S on her own. It remains fixed as exposure varies on a given menu. Let $p(\cdot | S; q) \in \Delta(S)$ denote the distribution of completed choices under menu S and exposure q . These probabilities can describe repeated choices by one consumer or choices by a population sharing the same menu and choice rule.

3.2 The Choice Rule

Let $\alpha \in (0, 1)$ be the consumer's *influence weight*. On each attempt, she draws from her baseline distribution with probability $1 - \alpha$ and from exposure with probability α . If the drawn alternative belongs to S , she selects it and the process ends. If it lies outside S , she makes another independent attempt using the same baseline, exposure, and influence weight. A baseline draw always completes the choice because the baseline is supported on S .

Extend $p_0(\cdot \mid S)$ by zero outside S , and let $M(\cdot \mid S; q)$ denote the distribution of the alternative drawn on an attempt:

$$M(\cdot \mid S; q) = (1 - \alpha)p_0(\cdot \mid S) + \alpha q.$$

The probability that an attempt completes the choice is $(1 - \alpha) + \alpha q_S$. It is strictly positive because $\alpha < 1$, so a choice is eventually completed with probability one. The completed-choice distribution is M conditional on S . For each $x \in S$,

$$p(x \mid S; q) = \frac{(1 - \alpha)p_0(x \mid S) + \alpha q(x)}{(1 - \alpha) + \alpha q_S}, \quad x \in S. \quad (1)$$

This choice rule defines *aspiration-weighted influence*. If $q_S = 0$, every exposure draw is unattainable and completed choices follow the baseline. If $q_S = 1$, every draw completes the choice and the rule is the mixture $(1 - \alpha)p_0 + \alpha q$.

The denominator in (1) is the completion probability on each attempt. The expected number of attempts is therefore $[(1 - \alpha) + \alpha q_S]^{-1}$. A lower attainable share increases this expected duration. It also reduces the fraction of completed choices that originate from exposure.

Figure 1 shows the construction. The distribution M combines baseline choice and exposure over $\mathcal{X}$. Conditioning M on S gives the distribution of completed choices.

For $q_S > 0$, conditioning exposure on S before mixing would give $(1 - \alpha)p_0(\cdot \mid S) + \alpha q(\cdot \mid S)$. In that rule, conditional exposure has weight α at every attainable share. It agrees with (1) exactly when $q_S = 1$ or $q(\cdot \mid S) = p_0(\cdot \mid S)$. Appendix B compares these rules with other ways to respond to an unattainable draw.

Across menus, let $u : \mathcal{X} \rightarrow \mathbb{R}_{++}$ be a positive index. A *Luce baseline* assigns each $x \in S$ probability proportional to $u(x)$, normalized by the sum over alternatives $y \in S$:

$$p_0(x \mid S) = \frac{u(x)}{\sum_{y \in S} u(y)}, \quad x \in S. \quad (2)$$

Combined with the same influence weight α on every menu, this baseline gives the *Aspiration-Weighted Luce Model*. Section 5.2 provides conditions for this cross-menu specification.

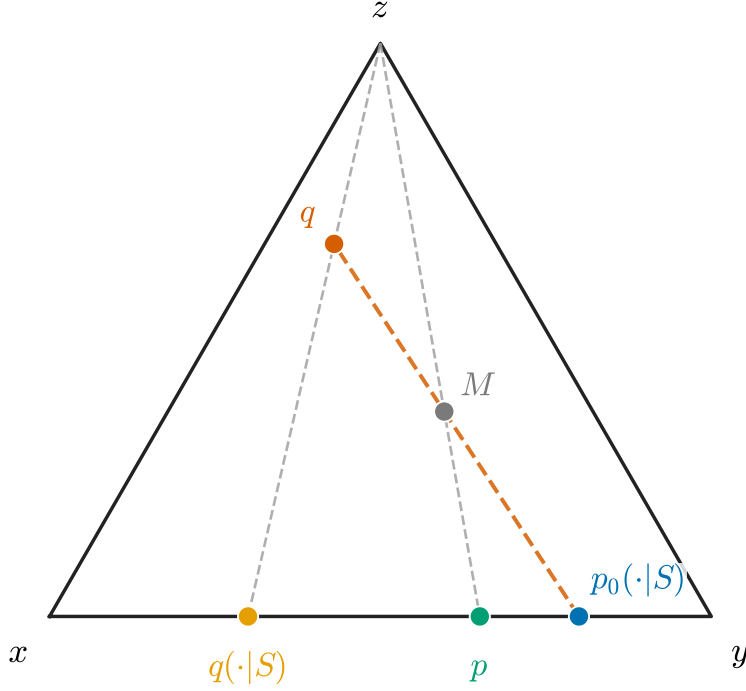


Figure 1: Aspiration-weighted influence. The attempt distribution M combines baseline choice with exposure. Conditioning M on the attainable menu produces the observed choice distribution p .

4 Comparative Statics

The model becomes transparent once attainable exposure is separated into composition and share. Let $\kappa = (1 - \alpha)/\alpha$. Dividing the numerator and denominator of (1) by α gives

$$p(\cdot | S; q) = \frac{\kappa p_0(\cdot | S) + q|_S}{\kappa + q_S}. \quad (3)$$

When $q_S > 0$, substitute $q|_S = q_S q(\cdot | S)$ to obtain

$$p(\cdot | S; q) = [1 - \beta(q_S)] p_0(\cdot | S) + \beta(q_S) q(\cdot | S), \quad \beta(\mu) = \frac{\alpha \mu}{(1 - \alpha) + \alpha \mu}. \quad (4)$$

The scalar $\beta(q_S)$ is the probability that the successful attempt came from the exposure branch. Whenever some exposure lies outside the menu, $\beta(q_S) < \alpha$. The consumer may be willing to follow exposure with probability α on every attempt, yet exposure appears less often among completed choices because some of its draws lead nowhere.

Equation (4) assigns a distinct role to each feature of attainable exposure. The conditional

distribution $q(\cdot \mid S)$ selects the endpoint toward which choice moves. The attainable share q_S determines the distance traveled toward that endpoint. Raising α or q_S increases β and moves choice farther from the baseline along the same segment.

Choice responds to exposure outside the menu only through the mass it absorbs. If two exposure distributions agree on S , they produce the same choice distribution even when they allocate the remaining mass to entirely different objects. Current choice therefore turns on two things: how much exposure is absorbed outside the menu, and the recommendation that remains inside it.

The paper's main comparative static holds the attainable recommendation fixed and changes only its share.

Proposition 4.1 (Aspirational dampening). *Fix S , $p_0(\cdot \mid S)$, and $\alpha \in (0, 1)$. Let q and q' have the same conditional exposure on S , with positive attainable shares. Then*

$$p(\cdot \mid S; q') - p(\cdot \mid S; q) = [\beta(q'_S) - \beta(q_S)] [q(\cdot \mid S) - p_0(\cdot \mid S)]. \quad (5)$$

If $q'_S < q_S$, choice under q' lies between $p_0(\cdot \mid S)$ and choice under q on the segment joining the baseline to their common conditional exposure.

A fall in the attainable share leaves the direction of influence intact and reduces its magnitude. Coordinate by coordinate,

$$p(x \mid S; q) - p_0(x \mid S) = \beta(q_S) [q(x \mid S) - p_0(x \mid S)].$$

Every gap from baseline contracts by the same factor. Additional exposure outside S therefore sends choice back along the same segment toward what the consumer would have chosen on her own, as Figure 2 shows. The attainable recommendation remains the destination of influence; only the distance traveled changes.

Dampening belongs to a wider class of rules. Any response that depends on the attainable share through a single increasing coefficient produces it, including the one-shot and finite-retry alternatives collected in Appendix B. Those rules differ in the shape of that coefficient. Section 5 identifies the schedule generated by persistent redrawing; Appendix B states the candidate class over which that protocol comparison is made.

Proposition 4.1 is a short-run comparative static. It holds while the menu and baseline remain stable. Over longer horizons, exposure may alter perceived quality, attention, income, or access,

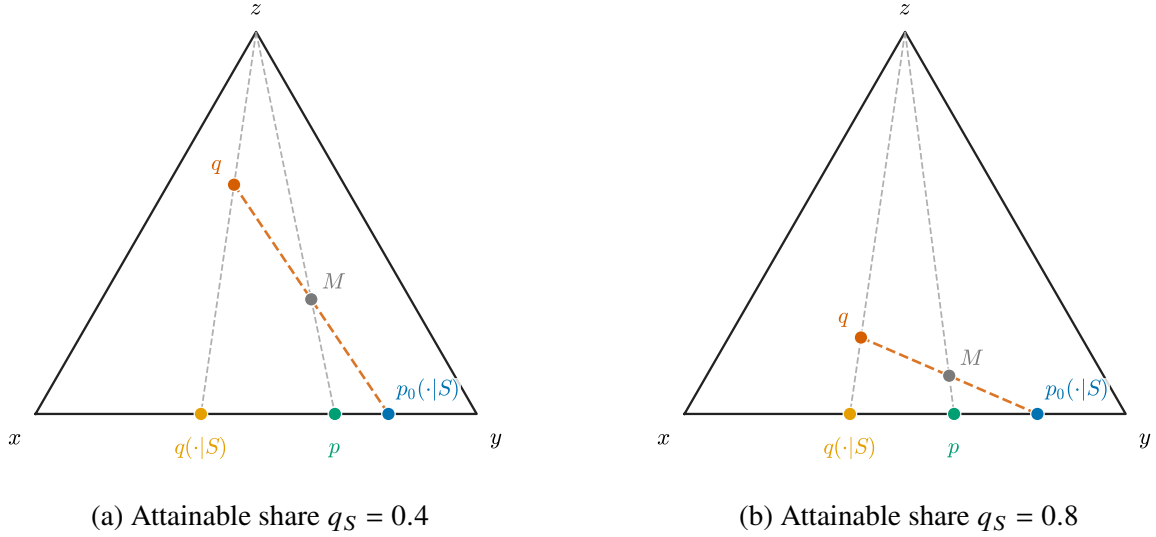


Figure 2: The attainable recommendation is the same in both panels. A larger attainable share places more realized weight on exposure and moves choice farther from the baseline.

and those adjustments can move the objects held fixed here. Within the current choice occasion, the model gives a sharp pair of predictions: aspirational identities leave choice unchanged, while aspirational mass moves choice along the line in (5) ¹.

The geometry in this section follows directly from the model. Recovering it from behavior alone, without assuming the repeated-sampling story in advance, is a different problem.

5 Characterization

We first characterize aspiration-weighted influence on a fixed menu through changes in exposure. A separate Luce extension asks when the menu-specific rules share an influence weight and a positive baseline index.

5.1 A fixed menu

Fix a proper menu $S \subseteq \mathcal{X}$ with $|S| \geq 3$, where $\mathcal{X}$ is the finite universe of alternatives. Retain $p(x \mid S; q)$ for the probability of choosing x and $q_S = \sum_{x \in S} q(x)$ for the attainable share of exposure.

¹This result distinguishes the mechanism from theories in which a particular unavailable alternative changes the evaluation of feasible options (Güney and Richter, 2015; Güney et al., 2018; Boissonnet and Ghersemgorin, 2026) where the characteristics or position of the unavailable alternative can redirect demand. Here the relevant object is the share of exposure that reaches the consumer’s menu. The analysis therefore focuses on a short-run choice problem in which the menu and the consumer’s baseline behavior remain fixed while exposure varies.

The observed rule is defined on $\mathcal{Q}_S = \{q \in \Delta(\mathcal{X}) : q_S > 0\}$, with $p(\cdot \mid S; q) \in \Delta(S)$. The domain includes zero exposure coordinates and profiles supported entirely on S . Let δ_x denote exposure concentrated on x . For distinct $a, b \in \mathcal{X}$, the profile $q + t(\delta_b - \delta_a)$ transfers t units of exposure from a to b , leaving every other coordinate fixed. It is feasible for $0 < t \leq q(a)$. The first two axioms apply to every profile and every feasible transfer in their stated domains.

We compare two kinds of transfers. The first reallocates exposure between attainable alternatives, keeping all unattainable exposure fixed. The second replaces unattainable exposure with exposure to an attainable alternative. Both comparisons keep the menu fixed and concern choices among alternatives whose own exposure has not changed. To illustrate, consider a photographer choosing among three affordable lenses x, y, k , while her feed also displays an unaffordable lens z .

Axiom 1 (Exposure Reallocation Independence). For every $q \in \mathcal{Q}_S$, $x \in S$, distinct $a, b \in S \setminus \{x\}$, and $0 < t \leq q(a)$,

$$p(x \mid S; q + t(\delta_b - \delta_a)) = p(x \mid S; q). \quad (6)$$

Holding an alternative's own exposure and the entire unattainable-exposure vector fixed, its choice probability is independent of how exposure is distributed among the other attainable alternatives. Any such reallocation can be made through finitely many of the stated transfers. Adding up makes the probability changes of the two affected alternatives equal and opposite. For the photographer, replacing some posts about lens y with posts about lens k leaves the probability of choosing x unchanged, with posts about x and every unaffordable lens held fixed.

For distinct $x, y \in S$, *relative choice* is the conditional frequency $p(x \mid S; q) / [p(x \mid S; q) + p(y \mid S; q)]$, when the denominator is positive. It describes choices from the original menu conditional on selecting a member of $\{x, y\}$; it neither changes the menu nor assumes a two-stage decision.

Axiom 2 (Relative Choice Independence). For every $q \in \mathcal{Q}_S$, distinct $x, y \in S$, $k \in S \setminus \{x, y\}$, $z \in \mathcal{X} \setminus S$, and $0 < t \leq q(z)$, let $q' = q + t(\delta_k - \delta_z)$. Then

$$p(x \mid S; q')p(y \mid S; q) = p(y \mid S; q')p(x \mid S; q). \quad (7)$$

Replacing unattainable exposure with exposure to k may change the probability of a choice in $\{x, y\}$. Whenever relative choice is defined at both profiles, the fraction of choices in the pair assigned to x is unchanged. The cross-product formulation accommodates zero choice probabilities. For the same photographer, replace posts about the unaffordable lens z with posts about k . If 60

percent of choices in $\{x, y\}$ were of x , that fraction remains 60 percent whenever the pair has positive choice probability under the new exposure.

The first two axioms also admit an exposure-insensitive rule. The next requires a positive but incomplete response in one comparison where all exposure is attainable.

Axiom 3 (Partial Responsiveness to Exposure). There exist $q \in \mathcal{Q}_S$ with $q_S = 1$, distinct $a, b \in S$, and $0 < t \leq q(a)$ such that

$$0 < p(b \mid S; q + t(\delta_b - \delta_a)) - p(b \mid S; q) < t. \quad (8)$$

The recipient gains choice probability, but by less than the exposure transferred to it. Only one such comparison is required. The upper bound is imposed at full attainable share, where raw and conditional exposure coincide.

Figure 3 pairs the exposure transfers with their restrictions on choice. With three attainable alternatives and one unattainable alternative, exposure lies in a tetrahedron and choice in a triangle.

Theorem 5.1 (Fixed-menu characterization). *Let $S \subseteq \mathcal{X}$ with $|S| \geq 3$. A choice rule on $\mathcal{Q}_S$ satisfies Axioms 1–3 if and only if there exist a unique $\alpha_S \in (0, 1)$ and a unique $p_0(\cdot \mid S) \in \Delta(S)$ such that*

$$p(x \mid S; q) = \frac{(1 - \alpha_S)p_0(x \mid S) + \alpha_S q(x)}{1 - \alpha_S + \alpha_S q_S} \quad (q \in \mathcal{Q}_S, x \in S). \quad (9)$$

No continuity or full-support assumption is required on $\mathcal{Q}_S$.

At a fixed vector of unattainable exposure, the first axiom and adding up imply an affine response to own exposure with the same slope for every attainable alternative. The second connects these coefficients across transfers into the menu. The third fixes the sign and range of the full-share response. Together they recover one baseline and one influence weight on this menu. The proof appears in Appendix A.2.

The common slope is a substantive restriction on the response to raw exposure frequencies. Unchanged own exposure need not imply unchanged choice: even product-specific scores depending only on own exposure can produce such a change through normalization. Appendix A.6 gives an example and separates the three axioms. The joint axioms also imply that $q|_S = q'|_S$ entails $p(\cdot \mid S; q) = p(\cdot \mid S; q')$. Independence from the allocation of unattainable exposure is therefore a conclusion, not a separate premise.

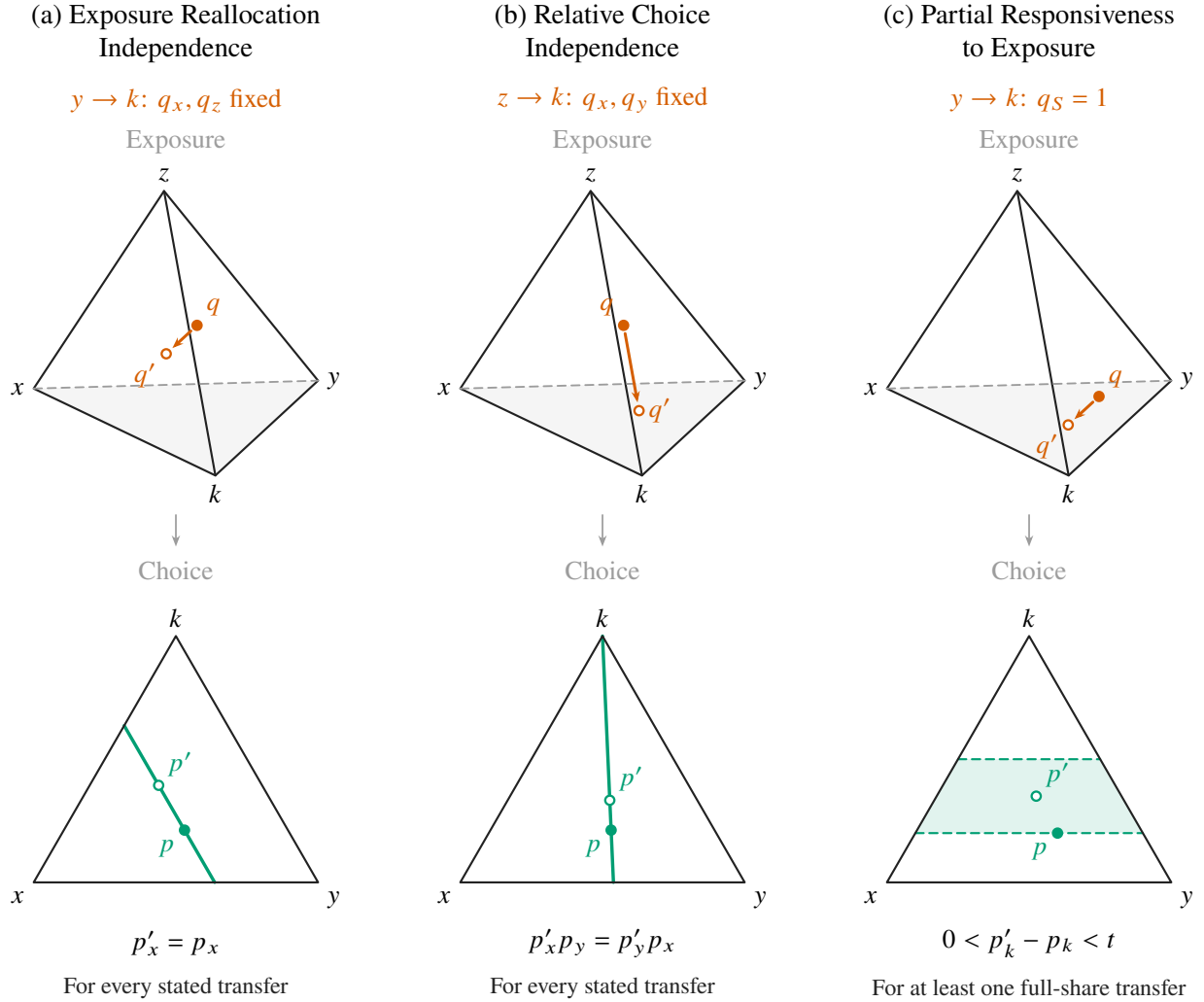


Figure 3: Exposure transfers and choice restrictions. The menu is $S = \{x, y, k\}$ and z is unattainable. Each upper panel shows an exposure transfer from q to q' ; the shaded face contains all exposure supported on S . The corresponding lower panel shows the displayed condition on $p' = p(\cdot \mid S; q')$, given the interior point $p = p(\cdot \mid S; q)$. In (a), $p'_x = p_x$, a line parallel to the yk edge. In (b), $p'_x p_y = p'_y p_x$, a line joining vertex k to the xy edge through p . In (c), the open band is $p_k < p'_k < p_k + t$; the dashed boundaries are excluded. The third axiom requires at least one such comparison at $q_S = 1$. Green lines and shading depict each condition separately for the illustrated comparison.

The recovered baseline $p_0(\cdot | S)$ is also the limit of choice as q_S tends to zero.² The subscript on α_S allows responsiveness to vary across menus.

5.2 A cross-menu Luce extension

The fixed-menu result leaves both baseline choice and responsiveness free to vary with the menu. This extension adds two comparisons that separate them: a difference under fully attainable exposure cancels the baseline term, while relative choice between two unexposed alternatives cancels the influence factor.

Let $\mathcal{S}$ be a family of proper menus, each with at least three alternatives, on which the fixed-menu axioms hold. For this extension only, impose *Full Support*: $p(x | S; q) > 0$ for every $S \in \mathcal{S}$, $q \in \mathcal{Q}_S$, and $x \in S$. Under Theorem 5.1, this is equivalent to positivity of every baseline coordinate.

Condition C1 (Exposure Response Consistency). For every $S, T \in \mathcal{S}$ and distinct $x, y \in S \cap T$,

$$p(x | S; \delta_x) - p(x | S; \delta_y) = p(x | T; \delta_x) - p(x | T; \delta_y). \quad (10)$$

Switching all exposure from y to x produces the same probability increment for x in either menu. Both exposure profiles are fully attainable on both menus, so the two differences equal α_S and α_T , respectively. The baseline cancels within each menu; only the probability increments are compared.

For the second comparison, keep $x, y \in S$ unexposed and concentrate exposure on $a \in S \setminus \{x, y\}$. Their probabilities are $(1 - \alpha_S)p_0(x | S)$ and $(1 - \alpha_S)p_0(y | S)$. Hence

$$\frac{p(x | S; \delta_a)}{p(x | S; \delta_a) + p(y | S; \delta_a)} = \frac{p_0(x | S)}{p_0(x | S) + p_0(y | S)}. \quad (11)$$

Both members of the pair are attainable but receive no exposure. Fully attainable exposure thus reveals their relative baseline choice.

Condition C2 (Baseline Relative Choice Consistency). For every $S, T \in \mathcal{S}$, distinct $x, y \in S \cap T$, $a \in S \setminus \{x, y\}$, and $b \in T \setminus \{x, y\}$,

$$\frac{p(x | S; \delta_a)}{p(x | S; \delta_a) + p(y | S; \delta_a)} = \frac{p(x | T; \delta_b)}{p(x | T; \delta_b) + p(y | T; \delta_b)}. \quad (12)$$

²If choice is separately specified at zero-share profiles, equating it with this limit requires a boundary condition, such as continuity there. See Appendix A.2.

When neither member of the pair receives exposure, their relative choice agrees across menus. The displayed alternatives a and b need not coincide; the menus need only share the compared pair for this condition to apply. The condition is equivalent to agreement of recovered baseline ratios. It places no restriction on influence weights, which cancel separately within each menu even when $\alpha_S \neq \alpha_T$.

Theorem 5.2 (Cross-menu characterization). *Let $|\mathcal{X}| \geq 4$, and let $\mathcal{S}$ be a family of proper menus, each with at least three alternatives. Suppose every three distinct alternatives in $\mathcal{X}$ lie in some member of $\mathcal{S}$. On each menu, let the rule satisfy Axioms 1–3 and Full Support. Conditions C1 and C2 hold if and only if there exist a common $\alpha \in (0, 1)$ and a positive index $u : \mathcal{X} \rightarrow \mathbb{R}_{++}$ such that*

$$p(x \mid S; q) = \frac{(1 - \alpha) \frac{u(x)}{\sum_{y \in S} u(y)} + \alpha q(x)}{1 - \alpha + \alpha q_S} \quad (S \in \mathcal{S}, q \in Q_S, x \in S). \quad (13)$$

The influence weight is unique, and u is unique up to a common positive factor.

This is the Aspiration-Weighted Luce Model. Triple coverage is a sufficient condition: it permits baseline ratios within each triple to be read from one menu and connects menu-specific influence weights through shared pairs. It is not necessary for every menu family. Appendix A.3 proves the result; Appendix A.6 distinguishes existence from identification under a maintained common index.

The Luce restriction concerns the baseline, not relative choice at arbitrary exposure profiles. For example, let $\mathcal{X} = \{1, 2, 3, 4, 5\}$, $u \equiv 1$, $\alpha = 1/2$, and $q = \delta_1$. Relative choice of 1 within $\{1, 2\}$ is $4/5$ on $S = \{1, 2, 3\}$ and $5/6$ on $T = \{1, 2, 3, 4\}$, although both rules have the same (α, u) . The unexposed-pair comparison isolates the baseline relation rather than imposing menu independence on the entire choice rule.

The theorem characterizes the observed menus and their positive-share domains. Predictions on other menus are model-based extensions. Full Support and the cross-menu restrictions are not assumptions of the fixed-menu identification or aggregate feasibility results.

6 Identification

The characterization studies the entire map from exposure to choice. Empirical work usually observes only a few exposure regimes. Two regimes recover the structural parameters when they describe the same consumer in the same choice environment.

6.1 Regime Stability

A *regime* is a pair (q, P) consisting of an exposure distribution and the choice distribution it induces for the same observational unit. The identification argument is algebraic, but its structural interpretation is conditional on the following invariance.

Assumption 6.1 (Exposure-regime stability). Across the regimes used for identification, the menu S , the baseline distribution $p_0(\cdot | S)$, and the influence weight α remain fixed. A regime affects choice only through the exposure distribution the consumer receives.

Random assignment can isolate a change in exposure. Repeated observations of the same consumer over a short window can make a stable baseline and influence weight more plausible. Common prices, inventory, geography, and account eligibility help hold the menu fixed; a common account tier addresses this menu component alone. A supply-side change in available content is useful only when it does not also change product quality, prices, or the composition of consumers being compared. None of these features delivers all three parts on its own.

The algebra itself does not require exposure to be statistically exogenous. It requires the same (S, p_0, α) to govern each observed pair (q_i, P_i) . Endogenous recommendations can satisfy the identities when they respond to variables that leave those objects unchanged. A causal interpretation of the exposure shift demands the stronger design argument appropriate to the application.

The model also yields restrictions that can reveal incompatibility with the maintained comparison, but they do not diagnose its source. If attainable exposure is unchanged while choice moves, the joint hypothesis of invariance to aspirational identities, parameter stability, a common population, and correctly measured exposure fails. If only the attainable share changes and choice leaves the line implied by Proposition 4.1, the same joint hypothesis fails. Either pattern may reflect a moving baseline, a changing influence weight, menu instability, sorting, exposure error, or a different behavioral rule. Passing these restrictions establishes compatibility with the model and the stability assumption; it does not establish stability independently of the research design.

6.2 A Single Regime

One regime blends baseline choice with influence. Fix an interior purchase distribution P and observed exposure q . For every sufficiently small $\alpha > 0$, the vector

$$p_0^{(\alpha)} = \frac{[(1 - \alpha) + \alpha q_S]P - \alpha q|_S}{1 - \alpha}$$

remains a probability distribution and reproduces P . The same observed regime therefore admits a continuum of nearby decompositions. Nonnegativity still bounds the admissible values of α ; the failure concerns joint point identification of the influence weight and the baseline.

A second regime creates the missing comparison. Figure 4 shows the observable movement before either structural object has been recovered.

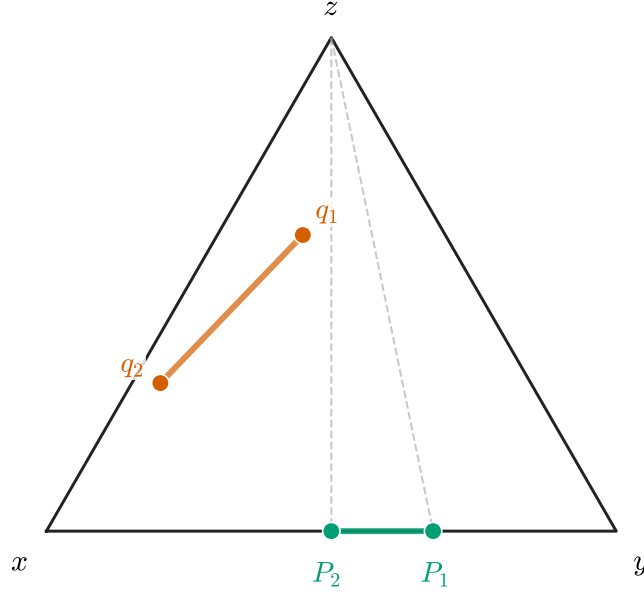


Figure 4: The identification problem. Exposure moves from q_1 to q_2 , while choice moves from P_1 to P_2 . The hidden baseline and the common influence weight must account for both movements.

6.3 Two Exposure Regimes

Fix a menu S with $|S| \geq 2$. In regime $i \in \{1, 2\}$, let q_i denote exposure and P_i the induced choice distribution. Write $v_i = q_i|_S$ for attainable exposure and $\mu_i = q_{i,S}$ for its total share. Before the final normalization, equation (1) becomes

$$\left[(1 - \alpha) + \alpha \mu_i \right] P_i = (1 - \alpha) p_0 + \alpha v_i. \quad (14)$$

The left side restores the probability mass removed when the attempt distribution was conditioned on S . Subtracting the two regime equations eliminates the common baseline.

Define the observed choice movement and the corresponding adjusted exposure movement by

$$\Delta P = P_2 - P_1, \quad A = (q_2 - q_1)|_S + (1 - \mu_2)P_2 - (1 - \mu_1)P_1. \quad (15)$$

Rearranging the differenced equation gives the central restriction

$$\Delta P = \alpha A. \quad (16)$$

The vector A combines the change in exposure on the menu with the change in the mass removed by normalization. The observed movement in choice must point in the same direction, and the common scalar is the influence weight.

Theorem 6.2 (Two-regime identification). *Fix a menu S with $|S| \geq 2$ and two regimes (q_1, P_1) and (q_2, P_2) on that menu, with $v_i = q_i|_S$, $\mu_i = q_{i,S}$, and ΔP and A as in (15). Suppose $A \neq 0$ and define*

$$\hat{\alpha} = \frac{A^\top \Delta P}{A^\top A}, \quad \hat{p}_0 = \frac{[(1 - \hat{\alpha}) + \hat{\alpha}\mu_1]P_1 - \hat{\alpha}v_1}{1 - \hat{\alpha}}. \quad (17)$$

Under Assumption 6.1, the two regimes are generated by aspiration-weighted influence if and only if

$$\Delta P = \hat{\alpha}A, \quad \hat{\alpha} \in (0, 1), \quad \hat{p}_0 \geq 0$$

componentwise. When these conditions hold, $(\hat{\alpha}, \hat{p}_0)$ is the unique pair of structural parameters that rationalizes both regimes.

The proportionality in equation (16) asks every coordinate to imply the same influence weight. Once that restriction holds, equation (17) reveals the baseline; the componentwise inequality checks that the recovered vector can describe choice. When $A = 0$ and $\Delta P \neq 0$, the two regimes reject the model. When $A = \Delta P = 0$, the exposure movement supplies no value for α , although nonnegativity of the implied baseline can still restrict its admissible range. A reallocation entirely among alternatives outside S falls into this last case: it may be large as a raw exposure change, yet it creates no identifying movement.

How much of the theorem tests the model, rather than inverting it, depends on the size of the menu. Both ΔP and A have coordinates summing to zero, so each lies in a space of dimension $|S| - 1$, and proportionality of two such vectors amounts to $|S| - 2$ restrictions. On a binary menu that count is zero: whenever $A \neq 0$, the scalar $\hat{\alpha}$ reproduces ΔP by construction, and only $\hat{\alpha} \in (0, 1)$

and $\widehat{p}_0 \geq 0$ retain content. Two regimes on a binary menu therefore recover the parameters under the maintained model without testing it. Directional tests require $|S| \geq 3$, and additional regimes supply overidentifying restrictions at any menu size.

The theorem is a statement about population distributions, and the informativeness of a design is the magnitude of A . The model shows exactly where that magnitude comes from. When the two regimes share one conditional exposure r and differ only in their attainable shares,

$$A = \frac{\beta(\mu_2) - \beta(\mu_1)}{\alpha} (r - p_0),$$

so the adjusted exposure movement factors into the share movement, acting through the response schedule, and the distance from the recommendation to the baseline. Both factors are needed. Exposure that recommends what the consumer would choose anyway yields $A = 0$ at every pair of shares. The model announces that degenerate case in the data, since $r = p_0$ forces observed choice to coincide with the observed attainable composition in both regimes. When instead the regimes share one attainable share μ and differ in composition, write $r_i = q_i(\cdot \mid S)$ for the two conditional exposures. Then $A = [\mu + (1 - \mu)\beta(\mu)](r_2 - r_1)$, and informativeness is the size of the compositional change.

Figure 5 makes the argument visible in the three-alternative case $\mathcal{X} = \{x, y, z\}$ with $S = \{x, y\}$. Conditioning any distribution on S moves it along a line through the outside vertex z to the edge xy . The attempt distribution $M_i = M(\cdot \mid S; q_i)$ of Section 3 must therefore lie on the line from z through the observed point P_i . A candidate baseline selects one point on each of these two conditioning lines. The same influence weight must work in both regimes, so $M_2 - M_1 = \alpha(q_2 - q_1)$: the segment joining the two attempt distributions must be parallel to the observed exposure shift. The baseline in the left panel satisfies this requirement. The candidate in the right panel reaches the observed choices after conditioning, but it implies different mixing weights across the two regimes.

When the two regimes have the same positive attainable share $\mu_1 = \mu_2 = \mu$, their normalizations agree and

$$P_2 - P_1 = \beta(\mu)[q_2(\cdot \mid S) - q_1(\cdot \mid S)].$$

The coefficient is the model response in (4). Under the maintained representation, a nonzero composition comparison at known μ already identifies $\alpha = \beta(\mu)/[\mu + (1 - \mu)\beta(\mu)]$; a second share is not required for homogeneous parameter identification.

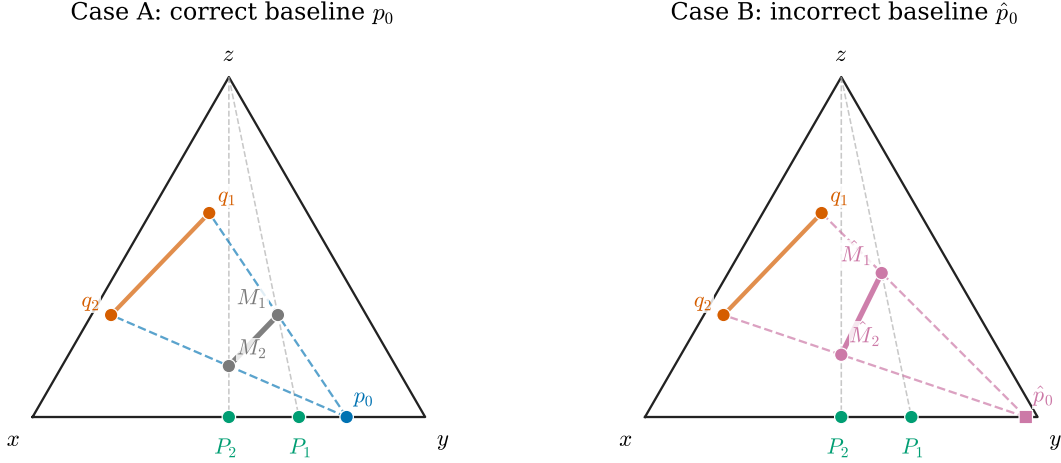


Figure 5: Recovering the hidden baseline. The correct candidate generates attempt distributions M_1 and M_2 whose difference is parallel to the exposure shift. The incorrect candidate fails the common- α restriction.

6.4 Additional Regimes

One informative pair fixes (α, p_0) at the population level. Every additional regime must satisfy equation (1) with those same parameters. More exposure variation therefore supplies overidentifying restrictions rather than new flexibility. Appendix C gives the finite-regime rationalizability conditions and establishes generic identification for designs with several regimes.

These are overidentifying restrictions for the joint hypothesis that the model is correctly specified, exposure is measured correctly, the same population is being compared, and (S, p_0, α) are stable. A rejection does not identify which component failed. Passing them leaves Assumption 6.1 as a maintained design condition rather than turning it into a conclusion from the choice data.

With sampled exposure and choice frequencies, the unnormalized equation (14) provides a direct set of moment conditions, and the choice of that equation matters. The identity $\Delta P = \alpha A$ holds in the population, but $\widehat{\Delta P}$ and $\widehat{A}$ are built from the same empirical frequencies, so regressing one on the other correlates the regressor with the error. Working from (14) avoids that correlation and turns estimation into standard minimum distance. Appendix F develops the estimator, specification tests, and confidence sets for weak adjusted exposure movements. It also allows the exposure distribution to be estimated from a separate sample of demonstrations, and treats the case in which the regime boundaries themselves must be located in the stream.

When structural parameters are already calibrated and exposure is only partly recorded, choice restricts attainable exposure to a one-dimensional set. Appendix C derives that set and shows when

one additional exposure statistic selects a unique point.

7 The Influencer's Problem

The analysis so far takes the exposure distribution q as part of the consumer's environment and asks how she responds to it. A natural next question would focus on how such exposure is curated. When q is chosen by an influencer whose payoff depends on what her followers eventually buy, what must the influencer know about her followers, and how does she use that information when deciding what to display?

The influencer chooses q through the products she consumes and displays. Every display corresponds to her actual consumption, so q records the distribution of demonstrated products rather than a collection of costless claims. Within this environment, she can allocate her content freely across the universe $\mathcal{X}$.

Her content generates two kinds of returns. When a follower's completed choice selects an attainable product, the influencer receives a product-specific monetary return, such as a promotional commission. Products outside the follower's current menu cannot generate a purchase on that occasion, but displaying them may strengthen the influencer's reputation, convey exclusivity, or enhance her status as a trend-setter.

Attainable and unattainable content therefore serve different purposes: the former can redirect purchases, while the latter can create direct display value.

The consumer's choice rule links these two sides of the problem. It maps the influencer's demonstrated content q into the follower's completed-choice distribution p , and therefore determines how much attainable content is required to steer consumption. The resulting trade-off depends on the follower's influence weight and baseline behavior. For the influencer's optimal content choice, however, the full baseline distribution is unnecessary: baseline behavior enters only through the expected monetary return generated by baseline choice.

To formalize the problem, fix a proper menu $S \subseteq \mathcal{X}$ and an audience with influence weight $\alpha \in (0, 1)$ and baseline distribution $p_0(\cdot \mid S) \in \Delta(S)$. Write $\kappa = (1 - \alpha)/\alpha$. The influencer chooses a single exposure distribution $q \in \Delta(\mathcal{X})$, and its attainable share is $\mu = q_S$. When $\mu > 0$, let $r = q(\cdot \mid S)$ denote the conditional exposure on the menu. The follower's influenced-choice

distribution is

$$p(\cdot \mid S; q) = [1 - \beta(\mu)]p_0(\cdot \mid S) + \beta(\mu)r, \quad \beta(\mu) = \frac{\mu}{\kappa + \mu}. \quad (18)$$

The expected number of attempts required to complete the decision is

$$\mathbb{E}[\tau \mid \mu] = \frac{1}{(1 - \alpha) + \alpha\mu} = \frac{1 + \kappa}{\kappa + \mu}.$$

At $\mu = 0$, every fresh attempt selects the baseline source with probability $1 - \alpha > 0$. Completion therefore occurs almost surely, with expected duration $1/(1 - \alpha)$. Fresh-source redrawing remains well defined when exposure is supported entirely outside S . A protocol that committed the consumer to keep drawing from q after selecting the exposure source would produce noncompletion at this boundary unless the environment specified a separate fallback or abandonment rule.

For each $x \in S$, let $c(x) \geq 0$ be the influencer's return when the completed decision selects x . If declining to purchase is feasible, it can be represented by an outside option $0 \in S$ with $c(0) = 0$. For each $z \in X \setminus S$, let $\phi(z) \geq 0$ be the direct display value of featuring z . This value may capture image, exclusivity, or audience-building benefits that accrue even though the current audience cannot select the displayed product.

All payoffs are measured per initiated decision occasion. The influencer allocates one unit of exposure capacity through q . Consumer redrawing changes the path to completed choice but leaves that capacity fixed. The expected completed-choice return is $c^\top p(\cdot \mid S; q)$, while the direct display value assigned to unattainable products is

$$\sum_{z \in X \setminus S} \phi(z)q(z).$$

Let $d \geq 0$ be the cost per attempt. The influencer solves

$$\max_{q \in \Delta(X)} \left\{ c^\top p(\cdot \mid S; q) + \sum_{z \in X \setminus S} \phi(z)q(z) - d \frac{1 + \kappa}{\kappa + q_S} \right\}. \quad (19)$$

Define

$$c_0 = c^\top p_0(\cdot \mid S), \quad G = \max_{x \in S} c(x) - c_0, \quad B = \max_{z \in X \setminus S} \phi(z), \quad (20)$$

and assume $G > 0$ and $B > 0$. The quantity G is the largest increase in completed-choice return

that attainable exposure can generate relative to the baseline. The quantity B is the highest direct display value available outside the menu.

Proposition 7.1 (Optimal content allocation). *Fix a proper menu $S \subsetneq \mathcal{X}$, an influence weight $\alpha \in (0, 1)$, a baseline distribution $p_0(\cdot \mid S) \in \Delta(S)$, completed-choice returns $c : S \rightarrow \mathbb{R}_+$, outside display values $\phi : \mathcal{X} \setminus S \rightarrow \mathbb{R}_+$, and a cost per attempt $d \geq 0$. Let $\kappa = (1 - \alpha)/\alpha$, define G and B by (20), and suppose $G > 0$ and $B > 0$. Every solution to (19) has the same attainable share*

$$\mu^*(d) = \min \left\{ 1, \max \left\{ 0, \sqrt{\frac{\kappa G + d(1 + \kappa)}{B}} - \kappa \right\} \right\}. \quad (21)$$

Whenever $\mu^(d) > 0$, every optimal conditional exposure on S assigns positive mass only to alternatives maximizing c . Whenever $\mu^*(d) < 1$, every optimal conditional exposure outside S assigns positive mass only to alternatives satisfying $\phi(z) = B$. Hence an optimal exposure distribution can be chosen with support on at most two alternatives: one attainable product with the highest completed-choice return and one unattainable product with the highest direct display value.*

When completion is costless,

$$\mu^*(0) = \min \left\{ 1, \max \left\{ 0, \sqrt{\kappa G/B} - \kappa \right\} \right\}. \quad (22)$$

As $\alpha \rightarrow 1$, $\mu^(0) \rightarrow 0$, whereas for $d > 0$,*

$$\mu^*(d) \longrightarrow \min \left\{ 1, \sqrt{d/B} \right\}. \quad (23)$$

Proof. Let $\mu = q_S$. When $\mu \in (0, 1)$, write

$$q|_S = \mu r, \quad q|_{\mathcal{X} \setminus S} = (1 - \mu)s,$$

where $r \in \Delta(S)$ and $s \in \Delta(\mathcal{X} \setminus S)$. At $\mu = 0$, the attainable composition is irrelevant; at $\mu = 1$, the outside composition is irrelevant. Equation (18) gives

$$c^\top p(\cdot \mid S; q) = c_0 + \beta(\mu)(c^\top r - c_0).$$

For every $\mu > 0$, $\beta(\mu) > 0$, so an optimal r is supported on $\arg \max_{x \in S} c(x)$. For every $\mu < 1$, the

direct display term is

$$(1 - \mu) \sum_{z \in \mathcal{X} \setminus S} \phi(z) s(z),$$

so an optimal s is supported on $\arg \max_{z \in \mathcal{X} \setminus S} \phi(z)$. Selecting one maximizer from each set produces an optimal exposure distribution with support on at most two alternatives.

After the two conditional compositions have been optimized, the objective equals the constant $c_0 + B$ plus

$$h_d(\mu) = G \frac{\mu}{\kappa + \mu} - B\mu - d \frac{1 + \kappa}{\kappa + \mu}.$$

Its derivatives are

$$h'_d(\mu) = \frac{\kappa G + d(1 + \kappa)}{(\kappa + \mu)^2} - B, \quad h''_d(\mu) = -\frac{2[\kappa G + d(1 + \kappa)]}{(\kappa + \mu)^3} < 0.$$

Thus h_d is strictly concave, and its unconstrained maximizer is

$$\sqrt{\frac{\kappa G + d(1 + \kappa)}{B}} - \kappa.$$

Projection onto $[0, 1]$ gives (21) and proves uniqueness of the attainable share. Setting $d = 0$ gives (22). Finally, $\alpha \rightarrow 1$ is equivalent to $\kappa \downarrow 0$. The costless expression converges to zero, while for $d > 0$ the projected expression converges to $\min \left\{ 1, \sqrt{d/B} \right\}$, which proves (23). $\square$

Only the scalar c_0 carries baseline behavior into the influencer's problem. It is the completed-choice return collected when no attainable content is displayed. The optimal conditional compositions do not depend on p_0 , and the attainable share depends on it only through $G = \max_x c(x) - c_0$. The influencer therefore needs the audience's influence weight and the single baseline statistic c_0 , rather than the full baseline distribution.

Conditional on the attainable share, reallocating exposure among products outside S leaves the consumer's completed-choice distribution unchanged. The influencer nevertheless assigns that exposure to products with the highest direct display value. Outside identities can therefore matter for the influencer's content allocation even though they have no additional effect on the consumer's current choice.

Raising μ increases completed-choice return through $G\mu/(\kappa + \mu)$, sacrifices $B\mu$ units of direct display value, and shortens expected completion. The return from steering choice and the saving in completion costs are concave in the attainable share, while the foregone display value is linear.

These three margins yield the unique share in Proposition 7.1.

When $d = 0$, the optimal attainable share is weakly single-peaked in the influence weight. It becomes positive once $\alpha > B/(B + G)$. Before projection onto $[0, 1]$, write

$$\tilde{\mu}(\kappa) = \sqrt{\kappa G/B} - \kappa.$$

Since

$$\tilde{\mu}'(\kappa) = \frac{1}{2} \sqrt{\frac{G}{B\kappa}} - 1,$$

the unconstrained share reaches its maximum at $\kappa = G/(4B)$, or equivalently at

$$\alpha = \frac{4B}{G + 4B}.$$

Projection preserves weak single-peakedness. If $G/B > 4$, the upper bound creates an interval on which all exposure is attainable.

The attainable share and the influencer's payoff follow different paths on the declining branch. Let $V^*(\alpha)$ denote the maximized value of (19) when $d = 0$. At an interior solution,

$$V^*(\alpha) = c_0 + B + (\sqrt{G} - \sqrt{\kappa B})^2.$$

This value rises strictly as α rises. At the lower corner, $V^*(\alpha) = c_0 + B$; at the upper corner, $V^*(\alpha) = c_0 + \alpha G$. The maximized payoff is therefore weakly increasing over the entire parameter range and strictly increasing whenever the optimal attainable share is positive. On the right side of Figure 6, a more responsive audience gives the influencer a higher payoff while requiring a smaller attainable share.

At an interior optimum with $d = 0$,

$$\mathbb{E}[\tau \mid \mu^*(0)] = \frac{1 + \kappa}{\sqrt{\kappa G/B}}.$$

As $\alpha \rightarrow 1$, expected completion time grows at rate $\kappa^{-1/2}$. A highly responsive audience can therefore be steered by a thin attainable share only through a longer path to completed choice. A positive cost per attempt prices that path and produces the positive high-influence limit in (23).

The exposure distribution represents one unit of feed capacity. The consumer's internal redraws change completion time but do not create additional posts. If direct display value instead accrued

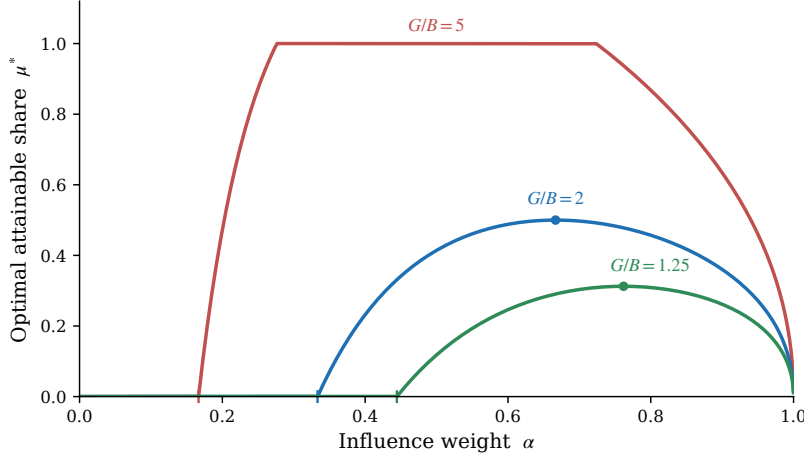


Figure 6: The influencer's optimal attainable share when completion is costless, for three values of G/B .

on every unattainable draw, the expected number of such draws would be $(1 - \mu)/(\kappa + \mu)$. After optimizing the conditional compositions, the payoff would become

$$\tilde{V}_d(\mu) = c_0 + \frac{(G - B)\mu + B - d(1 + \kappa)}{\kappa + \mu},$$

with derivative

$$\tilde{V}'_d(\mu) = \frac{\kappa G + (1 + \kappa)(d - B)}{(\kappa + \mu)^2}.$$

The derivative has a constant sign over μ . Encounter-based accounting therefore yields $\mu^* = 1$ when $\kappa G + (1 + \kappa)(d - B) > 0$ and $\mu^* = 0$ when this expression is negative; every share is optimal at equality. The solution is generically a corner, so the interior allocation and the single-peaked attainable-share curve disappear. Capacity-based accounting corresponds to an influencer allocating a fixed feed budget, while consumer redrawing governs the time required to convert that exposure into a completed choice.

Remark 7.2 (Item-specific display values). Suppose ϕ is defined on all of $\mathcal{X}$, so attainable products carry direct display value as well as completed-choice return. The direct display term in (19) becomes $\sum_{x \in \mathcal{X}} \phi(x)q(x)$. For a fixed $\mu \in (0, 1]$, the terms that depend on the attainable composition are

$$\beta(\mu) \sum_{x \in \mathcal{S}} c(x)r(x) + \mu \sum_{x \in \mathcal{S}} \phi(x)r(x) = \mu \sum_{x \in \mathcal{S}} \left[\phi(x) + \frac{c(x)}{\kappa + \mu} \right] r(x).$$

An optimal attainable composition can therefore be concentrated on an alternative

$$x_\mu \in \arg \max_{x \in S} \left\{ \phi(x) + \frac{c(x)}{\kappa + \mu} \right\}.$$

Whenever $\mu < 1$, unattainable exposure can be concentrated on an alternative maximizing $\phi(z)$ over $z \in \mathcal{X} \setminus S$. Recalling B from (20) as the maximum over $\mathcal{X} \setminus S$, the remaining choice is the one-dimensional problem

$$\max_{\mu \in [0,1]} \left\{ c_0 + (1 - \mu)B - d \frac{1 + \kappa}{\kappa + \mu} + \mu \max_{x \in S} \left[\phi(x) + \frac{c(x) - c_0}{\kappa + \mu} \right] \right\}.$$

An optimal exposure distribution can again be chosen with support on at most two alternatives. The attainable product may change with μ because direct display value and completed-choice return receive different relative weights as the attainable share changes. At $\mu = 0$, the attainable composition is irrelevant. Setting $\phi(x) = 0$ for every $x \in S$ recovers Proposition 7.1 and its closed-form solution.

8 Conclusion

Exposure can affect current choice even when some of the displayed alternatives are unattainable. In the model, choice depends on the composition and share of attainable exposure. Holding that composition fixed, a lower attainable share increases the weight on the consumer's baseline. Reallocating exposure among unattainable alternatives leaves choice unchanged.

For menus with at least three attainable alternatives and at least one unattainable alternative, three restrictions on exposure responses characterize the choice rule and recover a unique baseline and influence weight. Under the model, two informative exposure regimes jointly identify these objects when the menu and parameters remain fixed. Additional cross-menu restrictions yield a common influence weight and a Luce baseline. The identification results give empirical value to recording the attainable share alongside the composition of attainable exposure.

The sender application connects these choice responses to the content consumers see. Product returns favor attainable content, while image value can support unattainable content. A cost of delay weakly increases the optimal attainable share. With a positive cost of delay, the optimal attainable share remains bounded away from zero as the influence weight approaches one.

References

- APPADURAI, A. (2004): “The Capacity to Aspire: Culture and the Terms of Recognition,” in *Culture and Public Action*, ed. by V. Rao and M. Walton, Stanford, CA: Stanford University Press, 59–84.
- ARIDOR, G., R. JIMÉNEZ-DURÁN, R. LEVY, AND L. SONG (2024): “The Economics of Social Media,” *Journal of Economic Literature*, 62, 1422–1474.
- AZRIELI, Y. AND J. REHBECK (2026): “Marginal Stochastic Choice,” *Theoretical Economics*, 21, 373–401.
- BAGWELL, L. S. AND B. D. BERNHEIM (1996): “Veblen Effects in a Theory of Conspicuous Consumption,” *American Economic Review*, 86, 349–373.
- BLUME, L. E., W. A. BROCK, S. N. DURLAUF, AND Y. M. IOANNIDES (2011): “Identification of Social Interactions,” in *Handbook of Social Economics*, ed. by J. Benhabib, A. Bisin, and M. O. Jackson, Amsterdam: North-Holland, vol. 1, 853–964.
- BLUME, L. E., W. A. BROCK, S. N. DURLAUF, AND R. JAYARAMAN (2015): “Linear Social Interactions Models,” *Journal of Political Economy*, 123, 444–496.
- BOISSONNET, N. AND A. GHERSENGORIN (2026): “Grabbing the Forbidden Fruit: Restriction-Sensitive Choice,” *Journal of Economic Behavior & Organization*, 248, 107653.
- BORAH, A. AND C. KOPS (2019): “Choice via Social Influence,” Working Paper 06, Ashoka University, Department of Economics.
- BOUCHER, V., M. RENDALL, P. USHCHEV, AND Y. ZENOU (2024): “Toward a General Theory of Peer Effects,” *Econometrica*, 92, 543–565.
- BRAMOULLÉ, Y., H. DJEBBARI, AND B. FORTIN (2020): “Peer Effects in Networks: A Survey,” *Annual Review of Economics*, 12, 603–629.
- BURSZTYN, L., F. EDERER, B. FERMAN, AND N. YUCHTMAN (2014): “Understanding Mechanisms Underlying Peer Effects: Evidence From a Field Experiment on Financial Decisions,” *Econometrica*, 82, 1273–1301.
- CHAMBERS, C. P., T. CUHADAROGLU, AND Y. MASATLIOGLU (2023): “Behavioral Influence,” *Journal of the European Economic Association*, 21, 135–166.

- CHAMBERS, C. P., Y. MASATLIOGLU, P. NATENZON, AND C. RAYMOND (2025): “Weighted Linear Discrete Choice,” *American Economic Review*, 115, 1226–1257.
- CHAMBERS, C. P., Y. MASATLIOGLU, AND C. TURANSICK (2024): “Correlated Choice,” *Theoretical Economics*, 19, 1087–1117.
- CUHADAROGLU, T. (2017): “Choosing on Influence,” *Theoretical Economics*, 12, 477–492.
- DALTON, P. S., S. GHOSAL, AND A. MANI (2016): “Poverty and Aspirations Failure,” *The Economic Journal*, 126, 165–188.
- FARQUHAR, P. H. AND A. R. PRATKANIS (1993): “Decision Structuring with Phantom Alternatives,” *Management Science*, 39, 1214–1226.
- FUDENBERG, D., R. IJIMA, AND T. STRZALECKI (2015): “Stochastic Choice and Revealed Perturbed Utility,” *Econometrica*, 83, 2371–2409.
- GENICOT, G. AND D. RAY (2017): “Aspirations and Inequality,” *Econometrica*, 85, 489–519.
- GUL, F. AND W. PESENDORFER (2006): “Random Expected Utility,” *Econometrica*, 74, 121–146.
- GÜNEY, B. AND M. RICHTER (2015): “An Experiment on Aspiration-Based Choice,” *Journal of Economic Behavior & Organization*, 119, 512–526.
- GÜNEY, B., M. RICHTER, AND M. TSUR (2018): “Aspiration-Based Choice,” *Journal of Economic Theory*, 176, 935–956.
- HOPKINS, E. AND T. KORNIENKO (2004): “Running to Keep in the Same Place: Consumer Choice as a Game of Status,” *American Economic Review*, 94, 1085–1107.
- KASHAEV, N. AND N. LAZZATI (2021): “Peer Effects in Random Consideration Sets,” arXiv preprint 1904.06742, arXiv.
- KASHAEV, N., N. LAZZATI, AND R. XIAO (2026): “Peer Effects in Consideration and Preferences,” *The Review of Economic Studies*, rdag059, corrected proof.
- KILLICK, R., P. FEARNHEAD, AND I. A. ECKLEY (2012): “Optimal Detection of Changepoints With a Linear Computational Cost,” *Journal of the American Statistical Association*, 107, 1590–1598.

- KŐSZEGI, B. AND M. RABIN (2006): “A Model of Reference-Dependent Preferences,” *The Quarterly Journal of Economics*, 121, 1133–1165.
- LUCE, R. D. (1959): *Individual Choice Behavior: A Theoretical Analysis*, New York: Wiley.
- MACCHERONI, F., M. MARINACCI, AND A. RUSTICHINI (2006): “Ambiguity Aversion, Robustness, and the Variational Representation of Preferences,” *Econometrica*, 74, 1447–1498.
- MANSKI, C. F. (1993): “Identification of Endogenous Social Effects: The Reflection Problem,” *The Review of Economic Studies*, 60, 531–542.
- MANZINI, P. AND M. MARIOTTI (2014): “Stochastic Choice and Consideration Sets,” *Econometrica*, 82, 1153–1176.
- MASATLIOGLU, Y., D. NAKAJIMA, AND E. Y. OZBAY (2012): “Revealed Attention,” *American Economic Review*, 102, 2183–2205.
- MATĚJKA, F. AND A. MCKAY (2015): “Rational Inattention to Discrete Choices: A New Foundation for the Multinomial Logit Model,” *American Economic Review*, 105, 272–298.
- McFADDEN, D. (1974): “Conditional Logit Analysis of Qualitative Choice Behavior,” in *Frontiers in Econometrics*, ed. by P. Zarembka, New York: Academic Press, 105–142.
- McMANUS, D. A. (1992): “How Common Is Identification in Parametric Models?” *Journal of Econometrics*, 53, 5–23.
- MOOKHERJEE, D., S. NAPEL, AND D. RAY (2010): “Aspirations, Segregation, and Occupational Choice,” *Journal of the European Economic Association*, 8, 139–168.
- SACERDOTE, B. (2001): “Peer Effects with Random Assignment: Results for Dartmouth Roommates,” *The Quarterly Journal of Economics*, 116, 681–704.
- STOCK, J. H. AND J. H. WRIGHT (2000): “GMM with Weak Identification,” *Econometrica*, 68, 1055–1096.
- SULEYMANOV, E. (2026): “A Revealed Preference Framework for AI Alignment,” ArXiv:2603.27868.
- VEBLEN, T. (1899): *The Theory of the Leisure Class: An Economic Study of Institutions*, New York: Macmillan.

VOLPE, O. (2025): “Discrete Choice with Generalized Social Interactions,” Working paper, Harvard University.

WANG, G., C. ZOU, AND G. YIN (2018): “Change-Point Detection in Multinomial Data with a Large Number of Categories,” *Annals of Statistics*, 46, 2020–2044.

A Proofs

A.1 Proof of Proposition 4.1

Proof. Let $r = q(\cdot \mid S) = q'(\cdot \mid S)$. Equation (4) gives

$$p(\cdot \mid S; q) = [1 - \beta(q_S)]p_0 + \beta(q_S)r$$

and the analogous expression for q' . Subtracting yields (5). Moreover,

$$\beta'(\mu) = \frac{\alpha(1 - \alpha)}{[(1 - \alpha) + \alpha\mu]^2} > 0.$$

Hence $q'_S < q_S$ implies $0 < \beta(q'_S) < \beta(q_S)$. Both choice distributions lie on the segment from p_0 to r , and the smaller coefficient places choice closer to p_0 . $\square$

A.2 Proof of Theorem 5.1

Fix a proper menu $S \subsetneq \mathcal{X}$ with $|S| \geq 3$. In this proof, w denotes the entire vector $q|_{\mathcal{X} \setminus S}$ and ranges over $\mathcal{W}_S = \{w \in \mathbb{R}_+^{\mathcal{X} \setminus S} : \sum_{z \notin S} w_z < 1\}$. Write $\mu(w) = 1 - \sum_{z \notin S} w_z$. When a unit vector δ_x appears in a coordinate space indexed by a subset of $\mathcal{X}$ containing x , it denotes the restriction of that vector to the indicated space.

Lemma A.1 (Response conditional on unattainable exposure). *Exposure Reallocation Independence holds if and only if there exist unique coefficient functions $a_x : \mathcal{W}_S \rightarrow \mathbb{R}$ for $x \in S$ and $h : \mathcal{W}_S \rightarrow \mathbb{R}$ such that*

$$p(x \mid S; q) = a_x(w) + h(w)q(x), \quad x \in S, \quad (24)$$

where $w = q|_{X \setminus S}$, and, at each w ,

$$\sum_{x \in S} a_x(w) = 1 - \mu(w)h(w), \quad a_x(w) \geq 0, \quad a_x(w) + \mu(w)h(w) \geq 0 \quad (x \in S). \quad (25)$$

The axiom imposes no additional restriction across different vectors w ; in particular, the slope need not be positive.

Proof. Fix w and abbreviate $\mu = \mu(w) > 0$. The vector $v = q|_S$ ranges over all nonnegative vectors summing to μ . For fixed v_x , any two allocations of the other coordinates can be connected by finitely many feasible transfers among $S \setminus \{x\}$: move mass from a coordinate above its target to one below its target. The axiom preserves choice of x at every step. Thus $p(x | S; q) = f_x(v_x)$ for a function $f_x : [0, \mu] \rightarrow [0, 1]$, and $\sum_{x \in S} f_x(v_x) = 1$ whenever $v \geq 0$ sums to μ .

Put $a_x = f_x(0)$ and $g_x(s) = f_x(s) - a_x$. For distinct x, y , choose $k \in S \setminus \{x, y\}$. Applying the adding-up equality to $s\delta_x + (\mu - s)\delta_k$ and $s\delta_y + (\mu - s)\delta_k$ gives $g_x(s) = g_y(s)$. Denote the common increment function by g . Applying the same equality to $s\delta_x + t\delta_y + (\mu - s - t)\delta_k$ and $(s + t)\delta_x + (\mu - s - t)\delta_k$ gives $g(s + t) = g(s) + g(t)$ for $s, t \geq 0$ with $s + t \leq \mu$. Also $|g(s)| \leq 1$ because each f_x takes values in $[0, 1]$.

For every positive integer n and $0 \leq s \leq \mu/n$, $|g(s)| = |g(ns)|/n \leq 1/n$. Hence g is continuous at zero. Additivity gives continuity elsewhere, and rational approximation yields $g(s) = hs$ for all $s \in [0, \mu]$. This proves (24) without assuming regularity of the observed rule. Adding up gives the first part of (25); evaluating v_x at zero and at μ gives the inequalities. Conversely, these inequalities make each affine coordinate nonnegative throughout $[0, \mu]$, and adding up makes the vector a distribution satisfying the axiom. Its values at zero and positive own exposure uniquely determine each intercept and the common slope. $\square$

Proof of Theorem 5.1. Apply Lemma A.1. At $w = 0$, the attainable share is one and an exposure transfer of size t changes its recipient's probability by $h(0)t$. Partial Responsiveness to Exposure therefore gives $\alpha_S := h(0) \in (0, 1)$. Also $a_x(0) \geq 0$ and $\sum_{x \in S} a_x(0) = 1 - \alpha_S$.

Fix w , an unattainable coordinate z with $w_z > 0$, and $d \in (0, w_z]$. Let $w' = w - d\delta_z$, $\mu = \mu(w)$, and $\mu' = \mu + d$; use primes for the coefficients at w' . For any $x \in S$, choose distinct $y, k \in S \setminus \{x\}$. Give x exposure zero, y exposure s , and k exposure $\mu - s$, for $s \in [0, \mu]$. Transfer d from z to k . Relative Choice Independence gives $a_x(a'_y + h's) = (a_y + hs)a'_x$. Subtracting the equalities at $s = 0$ and $s = \mu > 0$ gives $a_x h' = h a'_x$. Summing over x and using (25) gives $(1 - \mu h)h' = h(1 - \mu' h')$.

Hence

$$a_x h' = h a'_x \quad (x \in S), \quad h - h' = dh h'. \quad (26)$$

These steps divide by neither a choice probability nor a slope.

Define $\kappa_S = (1 - \alpha_S)/\alpha_S > 0$ and $p_0(x | S) = a_x(0)/(1 - \alpha_S)$. This baseline lies in $\Delta(S)$. At $w = 0$, $h(0) = 1/(\kappa_S + 1)$ and $a_x(0) = \kappa_S p_0(x | S)/(\kappa_S + 1)$. From any w , remove its positive coordinates one at a time to obtain a finite path to zero, and work backwards along this path. If at its next point $h' = 1/(\kappa_S + \mu')$ and $a'_x = \kappa_S p_0(x | S)/(\kappa_S + \mu')$, then (26) gives $h(1 - dh') = h'$, with $1 - dh' = (\kappa_S + \mu)/(\kappa_S + \mu') > 0$. Thus $h = 1/(\kappa_S + \mu) > 0$, and the first identity gives $a_x = \kappa_S p_0(x | S)/(\kappa_S + \mu)$. Induction establishes these formulas at every w , independently of its allocation. Substituting them in (24) yields (9).

Conversely, a within-menu transfer in (9) leaves the denominator and all unaltered numerators fixed. A transfer from an unattainable alternative to k changes the common denominator and only k 's numerator, so (7) holds by cross multiplication. At full attainable share the recipient gains α_{St} ; taking $q = \delta_a$, $b \neq a$, and $t = 1$ verifies the existential response condition. Finally, the unique full-share slope $h(0)$ identifies α_S , and the full-share intercepts identify $p_0(\cdot | S)$ uniquely. $\square$

Zero-share choices. If $q_{n,S} \rightarrow 0$, each $q_n(x)$ for $x \in S$ tends to zero, so the represented choice tends to $p_0(\cdot | S)$. For a separately specified zero-share profile q^0 , the profiles $q_n = (1 - 1/n)q^0 + (1/n)\delta_x$, $x \in S$, converge to it from $\mathcal{Q}_S$. Continuity at q^0 therefore forces its choice to be $p_0(\cdot | S)$. Assigning that value at every zero-share profile gives the unique extension continuous there; the positive-domain axioms alone leave those actual values unrestricted.

A.3 Proof of Theorem 5.2

Lemma A.2 (The observable cross-menu comparisons). *Suppose the fixed-menu representation holds on every observed menu. For distinct $x, y \in S$,*

$$p(x | S; \delta_x) - p(x | S; \delta_y) = \alpha_S. \quad (27)$$

Under Full Support, for $a \in S \setminus \{x, y\}$,

$$\frac{p(x | S; \delta_a)}{p(x | S; \delta_a) + p(y | S; \delta_a)} = \frac{p_0(x | S)}{p_0(x | S) + p_0(y | S)}. \quad (28)$$

Consequently, Condition C1 equates influence weights of menus sharing a pair, and Condition C2 equates their baseline ratios for that pair, without requiring common influence.

Proof. All the displayed exposures are fully attainable, so the denominator in (9) is one. Subtraction gives (27). In (28), x and y are unexposed, and their choice probabilities are $(1 - \alpha_S)p_0(x | S)$ and $(1 - \alpha_S)p_0(y | S)$. The common positive factor cancels. For positive coordinates, equality of these conditional frequencies is equivalent by cross multiplication to equality of the ratios of the baseline coordinates. $\square$

Proof of Theorem 5.2. If a common (α, u) gives (13), each baseline is strictly positive. The fixed-menu theorem gives the three axioms, and baseline positivity gives Full Support. Lemma A.2 gives both cross-menu conditions, proving necessity.

Conversely, the fixed-menu theorem recovers $(\alpha_S, p_0(\cdot | S))$ on each menu. Full Support makes every recovered baseline coordinate positive: a zero coordinate at x would give $p(x | S; \delta_y) = 0$ for $y \neq x$ in S .

Every pair $\{x, y\}$ lies in an observed menu, because it can be extended to a triple. By Condition C1 and (27), all menus containing that pair have the same influence weight; denote this temporary value by α_{xy} . A menu covering a triple makes all three of its pair values equal. Two pairs with a common alternative therefore have equal values. Disjoint pairs $\{x, y\}$ and $\{a, b\}$ are connected through $\{x, a\}$, using the triples $\{x, y, a\}$ and $\{x, a, b\}$. Hence every pair value, and thus every menu's influence weight, equals one $\alpha \in (0, 1)$.

For distinct x, y , define the positive ratio $\omega(x, y) = p_0(x | S)/p_0(y | S)$ using any menu containing the pair, and set $\omega(x, x) = 1$. Condition C2 and Lemma A.2 make this definition independent of the selected menu. For distinct x, y, k , read the ratios from a menu containing all three to obtain

$$\omega(x, y)\omega(y, k) = \omega(x, k). \quad (29)$$

The identity extends to repeated indices because $\omega(y, x) = 1/\omega(x, y)$. Fix $x_0 \in X$, normalize $u(x_0) = 1$, and set $u(x) = \omega(x, x_0)$ for $x \neq x_0$. Then (29) gives $\omega(x, y) = u(x)/u(y)$. On each menu, the baseline has the same ratios as u and sums to one, so $p_0(x | S) = u(x)/\sum_{y \in S} u(y)$. Substitution yields (13).

The response difference (27) identifies α . Every pair is observed, so any other positive index generating the family has all the same ratios and differs from u by one common positive factor. This proves uniqueness. $\square$

A.4 Proof of Theorem 6.2

Proof. Under the model, equation (14) holds in both regimes. Subtracting regime 1 from regime 2 and collecting terms gives

$$(1 - \alpha)(P_2 - P_1) = \alpha[(v_2 - v_1) - s_2P_2 + s_1P_1].$$

Adding $\alpha(P_2 - P_1)$ to both sides yields equation (16). When $A \neq 0$, taking its inner product with A shows that any rationalizing influence weight must equal $\hat{\alpha}$ in equation (17). Rearranging the regime-1 equation then gives the stated expression for $\hat{p}_0$. Necessity of all three conditions follows.

For sufficiency, suppose the conditions in the theorem hold. Reversing the algebra that leads to equation (16) gives

$$[(1 - \hat{\alpha}) + \hat{\alpha}\mu_2]P_2 - \hat{\alpha}v_2 = [(1 - \hat{\alpha}) + \hat{\alpha}\mu_1]P_1 - \hat{\alpha}v_1.$$

By equation (17), the right side equals $(1 - \hat{\alpha})\hat{p}_0$, so the same pre-normalization equation holds in both regimes. Moreover,

$$(1 - \hat{\alpha})\mathbf{1}^\top \hat{p}_0 = (1 - \hat{\alpha}) + \hat{\alpha}\mu_1 - \hat{\alpha}\mu_1 = 1 - \hat{\alpha}.$$

Thus $\hat{p}_0$ sums to one. Componentwise nonnegativity places it in $\Delta(S)$, and equation (1) then reproduces both choice distributions. Finally, $A \neq 0$ makes $\hat{\alpha}$ the only scalar satisfying equation (16); the regime equation then makes $\hat{p}_0$ unique. $\square$

A.5 Proof of Proposition 7.1

The proof, including the case of positive completion costs, is given immediately after Proposition 7.1 in Section 7.

A.6 Examples and scope of the characterization

Separate content of the fixed-menu axioms. A constant rule $p(\cdot \mid S; q) = p_0$ satisfies both independence axioms but not Partial Responsiveness to Exposure. The rule $p(x \mid S; q) = e^{q(x)} / \sum_{y \in S} e^{q(y)}$ satisfies Relative Choice Independence: the compared probabilities share a denominator, and their numerators are unchanged. The transfer $\delta_a \rightarrow \delta_b$ at full share increases

the recipient's probability by $(e - 1)/(e + |S| - 1) \in (0, 1)$, so partial responsiveness also holds. However, for distinct $a, b, k \in S$, compare $q = (\delta_a + \delta_b)/2$ with $q' = q + t(\delta_b - \delta_a)$, $0 < t < 1/2$. The numerator for k is unchanged while the denominator increases, because $e^{1/2-t} + e^{1/2+t} > 2e^{1/2}$. Choice of k decreases, violating Exposure Reallocation Independence.

For the remaining separation, take any $|S| = m \geq 3$, a uniform baseline, and $\alpha = 1/2$. The one-shot rule $p(\cdot | S; q) = (1 - \alpha q_S)p_0 + \alpha q|_S$ satisfies Exposure Reallocation Independence and Partial Responsiveness to Exposure. For distinct $x, y, k \in S$ and $z \notin S$, let $q = \frac{1}{4}\delta_x + \frac{1}{4}\delta_k + \frac{1}{2}\delta_z$ and $q' = q + \frac{1}{4}(\delta_k - \delta_z)$. Relative choice of x within $\{x, y\}$ changes from $(m + 6)/(m + 12)$ to $(m + 5)/(m + 10)$; these differ for every $m \geq 3$. At $m = 3$ they are $3/5$ and $8/13$. Thus Relative Choice Independence fails. These examples establish the separate content of the three axioms on every fixed proper menu of the stated size.

The response to raw exposure frequencies. A product-specific exposure coefficient can change choice of an unaltered alternative without any direct cross-product term in its score. Consider

$$p^\eta(x | S; q) = \frac{\kappa p_0(x | S) + \eta_x q(x)}{\kappa + \sum_{y \in S} \eta_y q(y)}, \quad \kappa > 0, \quad p_0(\cdot | S) \in \Delta(S), \quad \eta_x > 0.$$

This rule satisfies Relative Choice Independence. Transferring t from attainable a to attainable b changes its denominator by $t(\eta_b - \eta_a)$. An unaltered alternative with positive numerator then changes choice probability if $\eta_a \neq \eta_b$. This example describes one way the first axiom can fail; it is not a characterization of all failures.

Zero probabilities and binary menus. If both pair totals are positive, the cross-product equality in Relative Choice Independence is equivalent to equality of relative choice. If both pair coordinates vanish at one profile, that equality alone says nothing about the pair at the other. The fixed-menu proof compares coefficients over the whole domain rather than dividing by a possibly zero choice probability. For $|S| = 2$, both independence axioms are vacuous. To see that partial responsiveness alone does not restore the representation, let $S = \{x, y\}$, $r = q(x)/q_S$, and set $p(x | S; q) = 1/4 + r/4 + r^2/4$ and $p(y | S; q) = 1 - p(x | S; q)$. At full share, the transfer $\delta_y \rightarrow \delta_x$ gives a probability increment of $1/2$, so the response condition holds. The rule is nonlinear in r on a fixed-share interval and therefore has no AWI representation. This example does not affect binary-menu identification under a maintained AWI rule.

Why the cross-menu condition concerns the baseline. Under the common representation, let $U_S = \sum_{k \in S} u(k)$ and $\kappa = (1 - \alpha)/\alpha$. For $x, y \in S$,

$$\frac{p(x \mid S; q)}{p(y \mid S; q)} = \frac{\kappa u(x) + U_S q(x)}{\kappa u(y) + U_S q(y)}.$$

Changing the menu changes U_S , so this ratio need not be menu-independent. Taking $q(x) = q(y) = 0$ removes that dependence and recovers the baseline ratio. For the example in the main text, the full choice vectors are $(2/3, 1/6, 1/6)$ on $\{1, 2, 3\}$ and $(5/8, 1/8, 1/8, 1/8)$ on $\{1, 2, 3, 4\}$. Their pair-conditional frequencies for option 1 within $\{1, 2\}$ are $4/5$ and $5/6$.

Pairwise agreement need not give a common index. Let $\mathcal{X} = \{1, \dots, 6\}$ and take $S_1 = \{1, 2, 3, 4\}$, $S_2 = \{3, 4, 5, 6\}$, and $S_3 = \{5, 6, 1, 2\}$. Assign the baseline $(2, 2, 1, 1)/6$ in the displayed order on each menu, and use one common influence weight. Each shared pair has relative baseline choice $1/2$ in both menus, so both cross-menu conditions hold. A common index would nevertheless require $u(1)/u(3) = u(3)/u(5) = u(5)/u(1) = 2$, whose product is 8 rather than 1. No positive index satisfies these requirements. The family covers every pair but not every triple; the example does not establish that triple coverage is necessary for every family.

What weaker menu overlap identifies. When existence of a common positive index is maintained, connect menus that share at least one alternative. Their within-menu ratios fix their relative normalizations along each such connection. A connected family therefore identifies the index up to one factor on its observed union. Disconnected components retain separate factors, and alternatives never observed remain unidentified. Without maintained existence, ratios obtained along different paths must also agree. Condition C1 instead uses a pair shared by two menus. On a weaker domain it equates influence weights only along chains of menus with intersections of size at least two. Single-alternative overlap cannot replace that requirement for this condition. Triple coverage in the theorem supplies both kinds of connection; no weakest-coverage claim is made.

Exposure domain and process interpretation. The axioms impose finite-transfer equalities throughout $\mathcal{Q}_S$; agreement at selected profiles does not verify them. On the full menu Axiom 2 is vacuous; on a singleton, Axiom 3 is unavailable. The latter's upper bound concerns only $q_S = 1$. Choice identifies a response rule, not a unique procedure; process comparisons require a specified class and completion convention.

B Alternative Stopping Rules

The main model checks availability after an exposure draw and repeats until some attempt lands on the menu. Other stopping rules preserve the direction selected by conditional exposure while changing the response to the attainable share. This appendix compares four homogeneous rules. The list is a candidate class rather than an exhaustive taxonomy. Section 5 characterizes the exposure–choice rule, not a unique stopping procedure: a mechanism generating the same rule remains observationally equivalent.

Let $\mu = q_S$ and $r = q(\cdot \mid S)$. Each rule below can be written as $p = [1 - b(\mu)]p_0 + b(\mu)r$, and they differ in the function b .

Stopping rule	What follows an outside draw	Response weight $b(\mu)$
Filter first	Remove outside exposure before selecting a source	α
One-shot fallback	An outside draw is followed by a baseline choice	$\alpha\mu$
At most N attempts	An outside draw permits another attempt, up to a fixed cap	$b_N(\mu)$ in (30)
Persistent redrawing	Continue until an attempt lands on the menu	$\frac{\alpha\mu}{1 - \alpha + \alpha\mu}$

Table 1: The attainable composition determines direction under all four rules; the stopping behavior determines response strength.

For the finite-cap rule, an exposure success on attempt t has probability $[\alpha(1 - \mu)]^{t-1}\alpha\mu$. Every path that remains after attempt N ends with baseline choice. The resulting response weight is

$$b_N(\mu) = \alpha\mu \sum_{t=0}^{N-1} [\alpha(1 - \mu)]^t = \frac{\alpha\mu}{1 - \alpha + \alpha\mu} [1 - \{\alpha(1 - \mu)\}^N]. \quad (30)$$

As N tends to infinity, $b_N(\mu)$ converges to the persistent-redrawing weight in the main text.

The capped rule needs a convention for occasions on which all N attempts land outside the menu. The schedule b_N assigns each such occasion to a baseline choice, so every occasion ends in a completed choice. Under an abandonment convention these occasions instead remain incomplete. The distribution conditional on completion then has exposure weight

$$\frac{b_N(\mu)}{1 - [\alpha(1 - \mu)]^N} = \frac{\alpha\mu}{1 - \alpha + \alpha\mu}$$

for every $N \geq 1$. Thus capped abandonment and persistent redrawing agree on the distribution of completed choices. A feasible no-purchase option in S is still a completed choice; conditioning further on positive purchases requires another normalization. This observation does not equate completion probabilities, durations, or type weights in pooled completion-selected samples.

On a proper menu $S \subsetneq \mathcal{X}$ with $|S| \geq 3$, all four rules satisfy Exposure Reallocation Independence and Partial Responsiveness to Exposure on the positive-share domain. They also leave the allocation of unattainable exposure irrelevant. Within this displayed homogeneous class, with failed capped occasions assigned to baseline choice, Relative Choice Independence is the remaining distinguishing restriction. At full attainable share every rule has response α . Theorem 5.1 then fixes the entire response schedule as

$$b(\mu) = \frac{\alpha\mu}{1 - \alpha + \alpha\mu}.$$

Filter-first, one-shot fallback, and every fixed finite retry cap differ from this schedule at some positive share and therefore cannot satisfy Relative Choice Independence on the whole exposure domain. Here b is only notation for a protocol's response weight, not an unrestricted object recovered by the new first two axioms. The comparison does not identify a unique process outside the stated class and completion convention.

The distinction also changes the sender application. Let $\mu_N^*(\alpha)$ maximize $Gb_N(\mu) - B\mu$ over $[0, 1]$.

Proposition B.1 (High-influence limit under a finite retry cap). *Fix a finite $N \geq 1$.*

- (i) *As $\alpha \rightarrow 1$, $b_N(\mu)$ converges uniformly on $[0, 1]$ to $1 - (1 - \mu)^N$.*
- (ii) *If $N = 1$, then $\mu_1^*(\alpha) \rightarrow 1$ when $G > B$ and $\mu_1^*(\alpha) \rightarrow 0$ when $G \leq B$.*
- (iii) *If $N \geq 2$, then*

$$\mu_N^*(\alpha) \rightarrow \begin{cases} 0, & GN \leq B, \\ 1 - \left(\frac{B}{GN}\right)^{1/(N-1)}, & GN > B. \end{cases}$$

Proof. The expression in (30) is continuous on the compact square $[0, 1]^2$ after setting its value at $\alpha = 1$ by continuity. At $\alpha = 1$,

$$b_N(\mu) = \mu \sum_{t=0}^{N-1} (1 - \mu)^t = 1 - (1 - \mu)^N.$$

Uniform continuity gives uniform convergence as $\alpha \rightarrow 1$. The sender objectives therefore converge

uniformly to

$$h_N^\infty(\mu) = G[1 - (1 - \mu)^N] - B\mu.$$

For $N = 1$, the finite- α objective is $(\alpha G - B)\mu$, which gives part (ii). For $N \geq 2$, h_N^∞ is strictly concave and

$$(h_N^\infty)'(\mu) = GN(1 - \mu)^{N-1} - B.$$

Its unique maximizer is the value in part (iii). Uniform convergence to an objective with a unique maximizer implies convergence of the finite- α maximizers. $\square$

Persistent redrawing sends the optimal attainable share toward zero as α approaches one. A fixed cap converges to a finite-contact problem and can retain a positive attainable share. This is the precise sense in which the declining right side of Figure 6 depends on persistent attempts.

C Further Identification Results

C.1 One and Many Regimes

The main text uses two regimes because that is the smallest comparison that can remove the common baseline.

Proposition C.1 (Single-regime nonidentification). *Fix a menu S , an exposure $q \in \Delta(X)$, and an interior choice distribution $P \in \Delta^\circ(S)$. There is an interval of influence weights $\alpha > 0$ and associated full-support baselines $p_0^{(\alpha)}$ for which aspiration-weighted influence maps q to P .*

Proof. For $\alpha \in (0, 1)$ define

$$p_0^{(\alpha)} = \frac{[(1 - \alpha) + \alpha q_S]P - \alpha q|_S}{1 - \alpha}.$$

The vector sums to one, and substitution into (1) reproduces P . As $\alpha \downarrow 0$, $p_0^{(\alpha)} \rightarrow P$ coordinatewise. Since P has full support, $p_0^{(\alpha)}$ remains strictly positive for every sufficiently small $\alpha > 0$. $\square$

For $K \geq 2$ regimes, let $\Delta_{ij} = P_j - P_i$ and

$$A_{ij} = (q_j - q_i)|_S + (1 - q_{j,S})P_j - (1 - q_{i,S})P_i.$$

Also define $D_i(\alpha) = (1 - \alpha) + \alpha q_{i,S}$ and $\psi_i(\alpha) = D_i(\alpha)P_i - \alpha q_i|_S$.

Proposition C.2 (Finite-regime rationalizability). *Let*

$$\mathcal{A} = \{\alpha \in (0, 1) : \Delta_{ij} = \alpha A_{ij} \text{ for every } i < j, \psi_i(\alpha) \in \mathbb{R}_+^S \text{ for some } i\}.$$

The K regimes are generated by common parameters (α, p_0) if and only if $\mathcal{A}$ is nonempty. For each $\alpha \in \mathcal{A}$, the pairwise restrictions imply $\psi_1(\alpha) = \dots = \psi_K(\alpha)$ and the associated baseline is $p_0 = \psi_i(\alpha)/(1 - \alpha)$. If $A_{ij} \neq 0$ for at least one pair, $\mathcal{A}$ contains at most one element.

Proof. Necessity follows by applying Theorem 6.2 to every pair and observing that the model gives $\psi_i(\alpha) = (1 - \alpha)p_0$ for every i .

For sufficiency, fix $\alpha \in \mathcal{A}$. The restriction $\Delta_{ij} = \alpha A_{ij}$ rearranges to

$$D_j(\alpha)P_j - D_i(\alpha)P_i = \alpha(q_j - q_i)|_S,$$

so $\psi_i(\alpha) = \psi_j(\alpha)$ for every pair. Let $\psi(\alpha)$ be their common value and set $p_0 = \psi(\alpha)/(1 - \alpha)$. Summing any $\psi_i(\alpha)$ over S gives $1 - \alpha$, and nonnegativity therefore places p_0 in $\Delta(S)$. By construction, every regime satisfies the unnormalized model equation.

Finally, if one A_{ij} is nonzero and both α and α' belong to $\mathcal{A}$, then $(\alpha - \alpha')A_{ij} = 0$, so $\alpha = \alpha'$. $\square$

C.2 Generic Identification

Identification is robust across exposure designs in the topological sense of [McManus \(1992\)](#). Fix a proper menu $S \subsetneq \mathcal{X}$ with $|S| = m \geq 2$ and view a profile $q = (q_1, \dots, q_K) \in \Delta(\mathcal{X})^K$ as the analyst's design. The model defines

$$\Phi_q : (0, 1) \times \Delta^\circ(S) \longrightarrow (\Delta(S))^K, \quad \Phi_q(\alpha, p_0) = (P_1, \dots, P_K).$$

Proposition C.3 (Generic identification with $K \geq 3$). *For $K \geq 3$, there is an open dense set $\mathcal{Q}^* \subset \Delta(\mathcal{X})^K$ such that, for every $q \in \mathcal{Q}^*$, the map Φ_q is injective and its Jacobian with respect to (α, p_0) has full column rank m at every $(\alpha, p_0) \in (0, 1) \times \Delta^\circ(S)$.*

Proof. For each regime write $v_i = q_i|_S$ and $\mu_i = \mathbf{1}^\top v_i$. Let $\mathcal{Q}^*$ contain the exposure profiles for which the vectors $\{v_i\}_{i=1}^K$ do not all lie on one affine line. Equivalently, some pair of differences $v_j - v_i$ and $v_k - v_i$ is linearly independent. A nonzero two-by-two minor witnesses this condition, so $\mathcal{Q}^*$ is open in the relative topology.

To see density, first perturb any profile into the relative interior of $\Delta(\mathcal{X})^K$. If the restricted vectors remain collinear and are not all equal, choose $v_j - v_i \neq 0$ and a third index k . Move an arbitrarily small amount of exposure in regime k from an outside alternative to a coordinate direction in S that is not parallel to $v_j - v_i$. The perturbed profile leaves the affine line. If all restricted vectors are equal, perturb two different regimes in two different coordinate directions. Since $m \geq 2$ and $K \geq 3$, the resulting differences are linearly independent. Thus $\mathcal{Q}^*$ is dense.

We next show that a profile in $\mathcal{Q}^*$ cannot generate the same choice distribution in every regime. If $P_i = P$ for all i , the unnormalized model equation gives

$$v_i - v_j = (\mu_i - s_j)P$$

for every pair, so all v_i lie on one affine line, a contradiction.

For the rank claim, the derivative of P_i in a baseline direction δ satisfying $\mathbf{1}^\top \delta = 0$ is

$$D_{p_0} P_i[\delta] = \frac{1 - \alpha}{(1 - \alpha) + \alpha \mu_i} \delta.$$

These directions contribute rank $m - 1$ in the stacked output. Differentiating with respect to α gives

$$\frac{\partial P_i}{\partial \alpha} = \frac{v_i - p_0 + (1 - \mu_i)P_i}{(1 - \alpha) + \alpha \mu_i}.$$

If the α direction belonged to the span of the baseline directions, there would be one tangent vector δ for which

$$v_i - p_0 + (1 - \mu_i)P_i = (1 - \alpha)\delta$$

for every i . Pairwise subtraction would give $A_{ij} = 0$ for all i, j . Theorem 6.2 would then imply $P_i = P_j$ for all pairs, contradicting the preceding paragraph. The Jacobian therefore has full rank m .

Finally, suppose two parameter pairs generate the same $(P_1, \dots, P_K)$. Because the choices do not all coincide, some pair has $P_i \neq P_j$. Theorem 6.2 identifies the same unique α from that pair under both parameterizations. The unnormalized equation then recovers the same p_0 . Hence Φ_q is injective. $\square$

C.3 Partially Observed Exposure

Suppose (α, p_0) is known and choice is observed, while exposure is missing or only partly recorded. Choice identifies attainable exposure up to its total share.

Proposition C.4 (The exposure set generated by one choice distribution). *Fix a proper menu $S \subsetneq \mathcal{X}$, $\alpha \in (0, 1)$, $\kappa = (1 - \alpha)/\alpha$, and $p_0 \in \Delta^\circ(S)$. If P is generated by some exposure, then $P \in \Delta^\circ(S)$ and the set of exposures generating P is*

$$\left\{ \tilde{q} \in \Delta(\mathcal{X}) : \tilde{q}|_S = (\kappa + \mu)P - \kappa p_0 \text{ for some } \mu \in [\underline{\mu}(P), 1] \right\},$$

where

$$\underline{\mu}(P) = \max \left\{ 0, \max_{x \in S} \kappa \left(\frac{p_0(x)}{P(x)} - 1 \right) \right\}.$$

For each admissible μ , the remaining mass $1 - \mu$ may be allocated arbitrarily over $\mathcal{X} \setminus S$.

Proof. Interiority follows from (1): $P(x) \geq (1 - \alpha)p_0(x) > 0$. If $\tilde{q}_S = \mu$, equation (3) yields

$$\tilde{q}|_S = (\kappa + \mu)P - \kappa p_0.$$

The right side sums to μ . Its x coordinate is nonnegative precisely when

$$\mu \geq \kappa \left(\frac{p_0(x)}{P(x)} - 1 \right).$$

Taking the maximum over coordinates and intersecting with $[0, 1]$ gives the stated interval. For a fixed attainable exposure vector, the represented rule is independent of how the residual mass is allocated outside S , so every such outside allocation is admissible. $\square$

The associated conditional exposure is

$$\tilde{q}(\cdot | S) = P + \frac{\kappa}{\mu}(P - p_0),$$

so changing μ moves along the ray from the baseline through observed choice.

Corollary C.5 (Choice restrictions with calibrated parameters). *For fixed (α, p_0) with a full-support baseline, a distribution $P \in \Delta(S)$ is generated by some exposure if and only if*

$$P(x) \geq (1 - \alpha)p_0(x) \quad \text{for every } x \in S.$$

Proof. Necessity follows from (1). For sufficiency set

$$q|_S = \frac{P - (1 - \alpha)p_0}{\alpha}.$$

The vector is nonnegative by hypothesis and sums to one, so it defines an exposure supported on S . Substitution returns P . $\square$

Proposition C.6 (One exposure statistic restores point identification). *Suppose $z = a^\top q|_S$ is observed for a known $a \in \mathbb{R}^S$ with $a^\top P \neq 0$. Then*

$$q_S = \frac{z - \kappa a^\top (P - p_0)}{a^\top P}, \quad q|_S = (\kappa + q_S)P - \kappa p_0.$$

Proof. Apply $a^\top$ to the identity $q|_S = \kappa(P - p_0) + q_S P$ and solve for q_S . $\square$

Direct observation of the attainable share corresponds to $a = \mathbf{1}$. Observation of one attainable exposure category corresponds to a coordinate vector and also determines the entire attainable exposure vector.

Corollary C.7 (Partially observed exposure). *Suppose the analyst observes $\{q(x) : x \in B\}$ for some $B \subseteq \mathcal{X}$.*

- (i) *If $B \cap S \neq \emptyset$, then $q|_S$ is point identified.*
- (ii) *If $B \subseteq \mathcal{X} \setminus S$, then the identified interval for q_S is $[\underline{\mu}(P), 1 - q(B)]$.*
- (iii) *If $B = \mathcal{X} \setminus S$, then $q_S = 1 - q(B)$ and $q|_S$ is point identified.*

Proof. Part (i) applies Proposition C.6 to any observed coordinate in S . For parts (ii) and (iii), the aspirational mass must satisfy $1 - q_S \geq q(B)$. A residual unobserved outside category permits every value up to $1 - q(B)$ in part (ii), while observing the whole complement forces equality in part (iii). $\square$

Example C.8 (A binary identified set). Let $S = \{x, y\}$, $p_0 = (0.4, 0.6)$, $\alpha = 1/2$, and $P = (0.6, 0.4)$. Then $\kappa = 1$, $\underline{\mu}(P) = 1/2$, and

$$q|_S(s) = (0.2 + 0.6s, 0.4s - 0.2), \quad \mu \in [1/2, 1].$$

At $s = 1/2$, attainable exposure is $(0.5, 0)$; at $s = 1$, it is $(0.8, 0.2)$. Both generate the same choice distribution. One observed attainable coordinate selects a unique point.

D Heterogeneous Influence Weights

The fixed-menu characterization describes a single exposure–choice rule. Under a common menu, common treatment exposure, and stable type weights, Exposure Reallocation Independence survives averaging individual AWI rules, whereas Relative Choice Independence need not survive heterogeneous influence weights. This appendix studies what the aggregate response reveals about that heterogeneity.

Fix a proper menu S . A consumer type is a pair $(\kappa, p_0) \in (0, \infty) \times \Delta(S)$, where $\kappa = (1 - \alpha)/\alpha$ is the parameter introduced in (3). Larger κ means a smaller influence weight. Types are distributed according to F , and F_κ denotes its marginal on the first coordinate. Assume

$$\text{supp } F_\kappa \subseteq [\underline{\kappa}, \bar{\kappa}], \quad 0 < \underline{\kappa} \leq \bar{\kappa} < \infty. \quad (31)$$

Every consumer in a treatment cell receives the same exposure, described by an attainable share $\mu \in (0, 1]$ and a conditional exposure $r \in \Delta(S)$. By (3), a type chooses according to

$$P_{\kappa, p_0}(\mu, r) = \frac{\kappa p_0 + \mu r}{\kappa + \mu}, \quad (32)$$

and the analyst observes the aggregate $\bar{P}(\mu, r) = \int P_{\kappa, p_0}(\mu, r) dF(\kappa, p_0)$.

D.1 Identification from Pooled Data

Proposition D.1 (Aggregate response and the distribution of influence weights). *Let F satisfy (31) and define the aggregate response schedule*

$$\bar{b}(\mu) = \int \frac{\mu}{\kappa + \mu} dF_\kappa(\kappa), \quad \mu \in (0, 1]. \quad (33)$$

(i) *For every $\mu \in (0, 1]$ and all $r, r' \in \Delta(S)$,*

$$\bar{P}(\mu, r) - \bar{P}(\mu, r') = \bar{b}(\mu) (r - r'), \quad (34)$$

and $\bar{b}' > 0$, $\bar{b}'' < 0$ on $(0, 1]$.

(ii) *Fix two distinct compositions $r^0, r^1 \in \Delta(S)$, and suppose $\bar{P}(\mu, r^0)$ and $\bar{P}(\mu, r^1)$ are observed for every μ in an open interval $I \subseteq (0, 1)$. Then F_κ is uniquely determined. The observed*

object is

$$H(\mu) \equiv \frac{\bar{b}(\mu)}{\mu} = \int \frac{1}{\kappa + \mu} dF_\kappa(\kappa), \quad (35)$$

the Stieltjes transform of F_κ .

(iii) The same observations determine $m_x(\kappa) \equiv \mathbb{E}[p_0(x) \mid \kappa]$ for every $x \in S$ and F_κ -almost every κ .

Part (i) is the reason pooling leaves anything behind at all. Changing the composition at a fixed share moves every type by the same multiple of $r - r'$, so the baseline terms cancel whatever their distribution, and two composition treatments deliver $\bar{b}$. Part (ii) then reads the whole distribution of influence weights off the curve $\bar{b}$, without a parametric family for F_κ . Part (iii) uses what remains after subtracting $\bar{b}(\mu)r$ to recover how baseline choice covaries with responsiveness.

D.2 Aggregation

Proposition D.2 (Aggregation). *Let F satisfy (31). Aggregate choice admits the representation*

$$\bar{P}(\mu, r) = \frac{\kappa_0 \bar{p}_0 + \mu r}{\kappa_0 + \mu} \quad (36)$$

for some $\kappa_0 > 0$ and $\bar{p}_0 \in \Delta(S)$, at every μ in an open subinterval of $(0, 1)$ and every r in a set containing two distinct compositions, if and only if F_κ is degenerate at κ_0 . When the representation holds, $\bar{p}_0 = \mathbb{E}[p_0]$.

With a common influence weight, heterogeneity in baseline choice aggregates into the mean baseline and the model survives intact. Heterogeneity in the influence weight changes the functional form of the response schedule, so pooled data no longer satisfy the fixed-menu representation of Theorem 5.1.

D.3 Testing the Response Schedule

Section 5 characterizes the exposure–choice rule of a homogeneous observational unit. With heterogeneous influence weights, pooled choice need not satisfy Relative Choice Independence even though each individual does. Failure of that condition in aggregate data can therefore reflect either a different individual rule or heterogeneity in responsiveness. It is not a failure of the new partial-response calibration merely because influence weights differ.

The heterogeneous persistent-redrawing class nevertheless has an exact reduced-form signature. Dividing the aggregate response by the attainable share gives

$$H(\mu) = \frac{\bar{b}(\mu)}{\mu} = \int \frac{1}{\kappa + \mu} dF_\kappa(\kappa).$$

Thus H is the Stieltjes transform of a probability distribution. For every integer $m \geq 1$,

$$(-1)^m H^{(m)}(\mu) = m! \int \frac{1}{(\kappa + \mu)^{m+1}} dF_\kappa(\kappa) > 0. \quad (37)$$

In particular, H is strictly decreasing. One-shot fallback has $H_1(\mu) = \alpha$, so it cannot be exactly observationally equivalent to a mixture of persistent redrawers at two distinct attainable shares. The gap is quantitative. Jensen's inequality gives $-H'(\mu) \geq H(\mu)^2$, and hence

$$\frac{1}{H(\mu_2)} - \frac{1}{H(\mu_1)} \geq \mu_2 - \mu_1 \quad (\mu_2 > \mu_1). \quad (38)$$

If $0 < \varepsilon < \alpha$ and a pooled persistent-redrawing schedule stays within ε of the constant α on an interval of length L , equation (38) requires

$$\varepsilon \geq \frac{\sqrt{1 + \alpha^2 L^2} - 1}{L}.$$

A positive one-shot response therefore cannot be approximated arbitrarily closely over a fixed nondegenerate share interval. More generally, a homogeneous rule with at most N attempts has

$$H_N(\mu) = \frac{b_N(\mu)}{\mu} = \alpha \sum_{t=0}^{N-1} [\alpha(1 - \mu)]^t,$$

a polynomial of degree $N - 1$. If H_N coincided with a pooled persistent-redrawing schedule on an open interval, their N th derivatives would coincide there. The derivative of H_N is zero at that order, whereas equation (37) is strictly signed. No finite retry cap, including one-shot fallback, therefore belongs exactly to the mixture class on an open interval.

Exact nonmembership does not imply strong discrimination in a finite design. Let b_∞ denote the homogeneous persistent-redrawing response. Equation (30) gives the uniform bound

$$0 \leq b_\infty(\mu) - b_N(\mu) = b_\infty(\mu)[\alpha(1 - \mu)]^N \leq \alpha^N, \quad \mu \in [0, 1].$$

Large retry caps can therefore be close to persistent redrawing throughout the observable share range. With finitely many share treatments, any numerical residual depends on the treatment grid, the norm used to measure fit, and sampling precision. Such an exercise can describe finite-design power; it cannot establish membership in, or exclusion from, the population mixture class.

The recovery result in Proposition D.1 is conditional in the same sense. If the observed H belongs to the Stieltjes class above, variation over an open interval identifies the unique distribution F_κ that rationalizes it within the persistent-redrawing mixture model. This does not validate the latent process type by type. An unrestricted mechanism that generates the same aggregate schedule remains observationally equivalent. If no probability distribution produces the observed H , pooled persistent redrawing is rejected. These are the bounds of the protocol interpretation.

D.4 Proofs

Proof of Proposition D.1. Integrating (32) over types gives

$$\bar{P}(\mu, r) = \int \frac{\kappa p_0}{\kappa + \mu} dF(\kappa, p_0) + \bar{b}(\mu) r, \quad (39)$$

where the first term does not depend on r . Subtracting the same expression at r' proves (34). On $[\underline{\kappa}, \bar{\kappa}]$ with $\underline{\kappa} > 0$, the integrand of (33) and its first two derivatives in μ are bounded uniformly, so differentiation passes under the integral:

$$\bar{b}'(\mu) = \int \frac{\kappa}{(\kappa + \mu)^2} dF_\kappa > 0, \quad \bar{b}''(\mu) = - \int \frac{2\kappa}{(\kappa + \mu)^3} dF_\kappa < 0.$$

This proves part (i).

For part (ii), choose $x \in S$ with $r^1(x) \neq r^0(x)$, which exists because the two compositions differ. Equation (34) gives

$$\bar{b}(\mu) = \frac{\bar{P}_x(\mu, r^1) - \bar{P}_x(\mu, r^0)}{r^1(x) - r^0(x)},$$

so $\bar{b}$, and hence $H = \bar{b}/\mu$, is observed on I . The support condition permits differentiation under the integral at every order and gives (37). Because F_κ is supported in $[\underline{\kappa}, \bar{\kappa}]$, the map $z \mapsto \int (\kappa + z)^{-1} dF_\kappa(\kappa)$ is analytic on the complement of $[-\bar{\kappa}, -\underline{\kappa}]$, and in particular on a complex neighborhood of $(-\underline{\kappa}, \infty)$. Its values on the open interval I therefore determine it everywhere on that domain by analytic continuation. A Stieltjes transform determines its measure, so F_κ is unique.

For part (iii), subtract $\bar{b}(\mu)r$ from (39) and read coordinate x :

$$R_x(\mu) \equiv \bar{P}_x(\mu, r) - \bar{b}(\mu)r(x) = \int \frac{\kappa m_x(\kappa)}{\kappa + \mu} dF_\kappa(\kappa).$$

Since m_x takes values in $[0, 1]$ and $\kappa \leq \bar{\kappa}$, the measure $\kappa m_x(\kappa) dF_\kappa(\kappa)$ is finite and positive, and R_x is its Stieltjes transform. The argument of part (ii) determines that measure from its values on I . Part (ii) already determines F_κ , and $\kappa \geq \underline{\kappa} > 0$, so m_x is determined F_κ -almost everywhere. $\square$

Proof of Proposition D.2. Suppose F_κ is degenerate at κ_0 . Then $\bar{b}(\mu) = \mu/(\kappa_0 + \mu)$, and the first term of (39) equals $\kappa_0 \mathbb{E}[p_0]/(\kappa_0 + \mu)$. Substituting both into (39) gives (36) with $\bar{p}_0 = \mathbb{E}[p_0]$.

Conversely, suppose (36) holds on an open subinterval I and at two distinct compositions. Applying part (i) of Proposition D.1 to the left side and differencing the right side at those two compositions gives $\bar{b}(\mu) = \mu/(\kappa_0 + \mu)$ for every $\mu \in I$, so $H(\mu) = (\kappa_0 + \mu)^{-1}$ there. The uniqueness established in part (ii) identifies F_κ with the point mass at κ_0 . Matching the terms in (39) that do not involve r then gives $\bar{p}_0 = \mathbb{E}[p_0]$. $\square$

E Sequential Exposure

The static model can be applied to a feed that is revealed one demonstration at a time. This appendix records the resulting path of choice when the consumer responds to the empirical distribution of what she has seen so far.

Fix S , $p_0 \in \Delta(S)$, and $\alpha \in (0, 1)$, and let $\kappa = (1 - \alpha)/\alpha$. Demonstrations arrive as a sequence $Z_1, Z_2, \dots$ in $\mathcal{X}$. After t observations, define

$$N_t(x) = \sum_{n=1}^t \mathbf{1}\{Z_n = x\}, \quad N_{t,S} = \sum_{x \in S} N_t(x), \quad \hat{q}_t = \frac{1}{t} \sum_{n=1}^t \delta_{Z_n}.$$

Applying equation (3) to $\hat{q}_t$ gives

$$P_t = \frac{\kappa p_0 + \hat{q}_t|_S}{\kappa + \hat{q}_{t,S}} = \frac{t\kappa p_0 + N_t|_S}{t\kappa + N_{t,S}}. \quad (40)$$

The structural parameters stay fixed; the changing object is the empirical exposure history.

Proposition E.1 (Updating after one more demonstration). *Let $w_t = t\kappa + N_{t,S}$.*

- (i) P_t depends on the history only through its category counts.

(ii) If $Z_{t+1} \notin S$, then

$$P_{t+1} = \frac{w_t P_t + \kappa p_0}{w_t + \kappa}.$$

Whenever $P_t \neq p_0$, the new demonstration moves choice strictly toward the baseline.

(iii) If $Z_{t+1} = x \in S$, then

$$P_{t+1} = \frac{w_t P_t + \kappa p_0 + \delta_x}{w_t + \kappa + 1}.$$

(iv) If the demonstrations are independent draws from a fixed distribution q , then $P_t \rightarrow p(\cdot \mid S; q)$ almost surely.

Proof. Part (i) follows from the count representation in (40). Since $w_t P_t = t\kappa p_0 + N_t|_S$, the numerator at date $t + 1$ is

$$w_t P_t + \kappa p_0 + (N_{t+1}|_S - N_t|_S).$$

The final increment equals zero when $Z_{t+1} \notin S$ and equals δ_x when $Z_{t+1} = x \in S$. The denominator rises by κ in the first case and by $\kappa + 1$ in the second, which proves parts (ii) and (iii). In part (ii), the weight on p_0 is $\kappa/(w_t + \kappa) \in (0, 1)$, so the movement is strict whenever $P_t \neq p_0$. For part (iv), the strong law gives $\widehat{q}_t \rightarrow q$ coordinatewise, and the map in equation (3) is continuous because its denominator is bounded below by $\kappa > 0$. $\square$

An unattainable demonstration changes no count inside the menu, yet it increases the amount of history relative to attainable exposure. Choice therefore retreats toward the baseline. An attainable demonstration adds one unit to the displayed alternative while the passage of one period adds κ units of baseline weight. The size of either update decreases with the length of the history.

This formulation treats the realized exposure stream as observed. A model in which the stream is sampled from latent regimes requires an additional statistical layer for regime detection; the choice implications above apply once the relevant exposure history has been specified.

F Estimation

The identification result is stated in population probabilities. This appendix develops a minimum-distance estimator when exposure and choice are observed through samples. The moment conditions preserve the unnormalized structure of equation (14), so the baseline enters linearly once the influence weight is fixed.

F.1 Moment Conditions

Fix a menu S with $m = |S| \geq 2$ and observe K exposure regimes. In regime i , let q_i be the exposure distribution, $v_i = q_i|_S$, $\mu_i = q_{i,S}$, and P_i the population choice distribution. A sample of N_i independent choices produces empirical frequencies $\widehat{P}_i$. Define $D_i(\alpha) = (1 - \alpha) + \alpha\mu_i$. Equation (14) suggests the sample moment

$$m_i(\alpha, p_0) = D_i(\alpha)\widehat{P}_i - (1 - \alpha)p_0 - \alpha v_i. \quad (41)$$

At the true parameters, $\mathbb{E}[m_i(\alpha_0, p_0)] = 0$ for every regime.

The coordinates of each moment sum to zero. Index the alternatives in S by $1, \dots, m$, and let

$$H = \begin{pmatrix} I_{m-1} & 0_{(m-1) \times 1} \end{pmatrix}$$

drop the last coordinate. Write $\widetilde{m}_i = Hm_i$, $\widetilde{p}_0 = Hp_0$, and stack the reduced moments into

$$\widetilde{m}(\theta) = (\widetilde{m}_1(\theta)^\top, \dots, \widetilde{m}_K(\theta)^\top)^\top, \quad \theta = (\alpha, \widetilde{p}_0).$$

There are $K(m - 1)$ moment conditions and m free parameters.

F.2 Concentrating Out the Baseline

For a positive-definite block-diagonal weight $W = \text{blkdiag}(W_1, \dots, W_K)$, consider

$$Q_N(\theta; W) = N \widetilde{m}(\theta)^\top W \widetilde{m}(\theta), \quad N = \sum_{i=1}^K N_i.$$

At a fixed α , the criterion is quadratic in $\widetilde{p}_0$. Its minimizer is

$$\widehat{\widetilde{p}}_0(\alpha) = \frac{1}{1 - \alpha} \left(\sum_{i=1}^K W_i \right)^{-1} \sum_{i=1}^K W_i H [D_i(\alpha)\widehat{P}_i - \alpha v_i]. \quad (42)$$

Estimation can therefore proceed through the one-dimensional concentrated criterion

$$Q_N^c(\alpha; W) = Q_N((\alpha, \widehat{\widetilde{p}}_0(\alpha)); W).$$

After minimizing over an interior interval for α , the last coordinate of p_0 is recovered from adding up. Imposing the simplex constraint in finite samples amounts to minimizing the same criterion over $(\alpha, p_0) \in [\varepsilon, 1 - \varepsilon] \times \Delta(S)$.

F.3 Sampling Variation and Weighting

Let $\pi_i = N_i/N$. Under multinomial sampling,

$$\text{Var}(\widehat{P}_i) = \frac{1}{N_i} \Sigma(P_i), \quad \Sigma(P_i) = \text{diag}(P_i) - P_i P_i^\top.$$

Hence

$$\Omega = \text{Var}(\sqrt{N} \widetilde{m}(\theta_0)) = \text{blkdiag} \left(\frac{D_1(\alpha_0)^2}{\pi_1} H \Sigma(P_1) H^\top, \dots, \frac{D_K(\alpha_0)^2}{\pi_K} H \Sigma(P_K) H^\top \right).$$

A two-step estimator first minimizes with a convenient preliminary weight, constructs a plug-in estimate $\widehat{\Omega}$, and then minimizes with $W = \widehat{\Omega}^{-1}$.

Let $g(\theta) = \mathbb{E}[\widetilde{m}(\theta)]$ and let G be its Jacobian at the true parameter. The block derivatives for regime i are

$$\frac{\partial \mathbb{E}[\widetilde{m}_i]}{\partial \widehat{p}_0^\top} = -(1 - \alpha) I_{m-1}, \quad \frac{\partial \mathbb{E}[\widetilde{m}_i]}{\partial \alpha} = H[(\mu_i - 1)P_i + p_0 - v_i].$$

Under the usual regularity and rank conditions,

$$\sqrt{N}(\widehat{\theta} - \theta_0) \Rightarrow \mathcal{N}\left(0, (G^\top \Omega^{-1} G)^{-1}\right).$$

The rank condition is the sample analogue of informative adjusted exposure variation in Section 6.

F.4 Estimated Exposure

Suppose regime i also contains T_i independent demonstrations drawn from q_i , producing empirical exposure $\widehat{q}_i$. Let $\widehat{v}_i = \widehat{q}_i|_S$, $\widehat{\mu}_i = \mathbf{1}^\top \widehat{v}_i$, and $\widehat{D}_i(\alpha) = (1 - \alpha) + \alpha \widehat{\mu}_i$. Replace (41) by

$$\widehat{m}_i(\alpha, p_0) = \widehat{D}_i(\alpha) \widehat{P}_i - (1 - \alpha) p_0 - \alpha \widehat{v}_i.$$

Independence of the exposure and choice samples gives the same population moment as before. The extra sampling variation changes the covariance matrix.

Write $\delta P_i = \widehat{P}_i - P_i$, $\delta v_i = \widehat{v}_i - v_i$, and $J_i = P_i \mathbf{1}^\top - I_m$. At the true parameter,

$$\widehat{m}_i(\theta_0) = D_i(\alpha_0)\delta P_i + \alpha_0 J_i \delta v_i + \alpha_0 (\mathbf{1}^\top \delta v_i) \delta P_i.$$

The final term is of smaller order. Since

$$\text{Var}(\delta v_i) = \frac{1}{T_i} \Sigma_i^q, \quad \Sigma_i^q = \text{diag}(v_i) - v_i v_i^\top,$$

the leading covariance of the unreduced moment is

$$\text{Var}(\widehat{m}_i(\theta_0)) = \frac{D_i(\alpha_0)^2}{N_i} \Sigma(P_i) + \frac{\alpha_0^2}{T_i} J_i \Sigma_i^q J_i^\top + O((N_i T_i)^{-1}).$$

Premultiplying and postmultiplying by H gives the block used in the optimal weight. When T_i/N_i grows without bound, the known-exposure covariance is recovered.

F.5 Locating Regimes in a Stream

The moments above take the partition of the data into regimes as given. A feed delivers exposure as a sequence, and the boundaries then have to be located before the structural moments can be applied. Suppose demonstrations are independent categorical draws with one break,

$$Z_t \sim q^{(1)} \text{ for } t \leq \tau, \quad Z_t \sim q^{(2)} \text{ for } t > \tau,$$

where $q^{(1)} \neq q^{(2)}$, the break date τ stays away from the endpoints, and both segment lengths diverge. A first stage estimates $(\widehat{\tau}, \widehat{q}^{(1)}, \widehat{q}^{(2)})$ by segmentation. Penalized-likelihood procedures locate an unknown number of breaks at linear computational cost (Killick et al., 2012), and change-point tests built for high-dimensional multinomial data apply directly to demonstration streams (Wang et al., 2018). The second stage feeds the estimated segments into the moment conditions above.

Break-dating error is negligible at first order. Suppose the estimator satisfies $\widehat{\tau} - \tau = O_p(1)$, as fixed-magnitude break estimators do. Replacing the true split by $\widehat{\tau}$ then misclassifies $O_p(1)$ observations, so the resulting error in each segment frequency is $O_p(T_i^{-1}) = o_p(T_i^{-1/2})$. The $\sqrt{T_i}$ exposure-frequency limit used in the previous subsection therefore remains the leading input to the

second stage.

A located break supplies exposure variation and nothing more. Interpreting the two segments through the model still requires Assumption 6.1. The baseline, the influence weight, and the menu must hold still across the break. The shifted stream must reach the consumer, and the choice window must line up with the break date. The segmentation stage reads the exposure stream alone, so it carries no information about whether the consumer moved at the same time.

F.6 Specification Tests and Weak Identification

The model is overidentified when $K(m - 1) > m$. With the optimal two-step weight,

$$J_N = Q_N(\hat{\theta}; \hat{\Omega}^{-1}) \Rightarrow \chi_{K(m-1)-m}^2$$

under correct specification. The test asks whether all regimes can be reconciled with one baseline and one influence weight.

When the adjusted exposure movements are small, the influence weight may be weakly identified even though the baseline remains well behaved at a fixed α . The concentrated criterion provides an identification-robust confidence set. For each candidate α , estimate the baseline by (42) using a covariance matrix valid under that candidate and retain the values satisfying

$$Q_N^c(\alpha; \hat{\Omega}(\alpha)^{-1}) \leq \chi_{(K-1)(m-1), 1-\gamma}^2.$$

At the true α , the minimized criterion tests the equality of the K regime-specific baseline moments and has $(K - 1)(m - 1)$ degrees of freedom. Inverting that test yields asymptotically valid coverage as the adjusted exposure movement approaches zero (Stock and Wright, 2000).

G A Relative-Entropy Representation

For $q_S > 0$, equation (3) can also be written as a distributional barycenter. Write $D_{\text{KL}}(r \parallel \pi) = \sum_{x \in S} r(x) \log[r(x)/\pi(x)]$, with the convention $0 \log 0 = 0$. Then

$$p(\cdot \mid S; q) = \arg \min_{\pi \in \Delta(S)} \{ \kappa D_{\text{KL}}(p_0(\cdot \mid S) \parallel \pi) + q_S D_{\text{KL}}(q(\cdot \mid S) \parallel \pi) \}.$$

Expanding the objective, the Lagrange conditions give $\pi(x) = [\kappa p_0(x \mid S) + q(x)]/(\kappa + q_S)$ for every $x \in S$, which is equation (3). Strict convexity on the relevant support gives uniqueness.

The representation views realized choice as a reverse-KL barycenter of baseline choice and conditional exposure. Their weights are κ and q_S , so the attainable share changes the importance of exposure without changing the distribution toward which it pulls. This connects the aggregation in the model to relative-entropy methods while preserving the revealed-choice interpretation of its weights ([Maccheroni et al., 2006](#)).